OFFICE OF THE REGISTRAR
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(ACADEMIC SECTION)

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No. UOM/Acad/25/349 Dated: 22-Sep-2025

CORRIGENDUM

It is notified for the information of all concerned that in this office's notification No. UOM/Acad/24/479 dated 05-Dec-2024 (copy enclosed), the word *"following"* appearing after the words *"[vide agenda item 24(3)], approved the"* shall be treated as deleted.

Further, after the words *"Academic Council made in its 22nd (Special) meeting held on 21-Oct-2024"* the phrase *"along with the following:"* shall be added.

This is issued with the approval of the Competent Authority.

Assistant Registrar Academics

Copy for information and necessary action (if any) to:

1. All Deans
2. Director QEC
3. Deputy Registrar Admissions
4. Chairperson, Department of CS & IT
5. Chairperson, Department of Artificial Intelligence
6. Chairperson, Department of Multimedia & Gamming
7. Incharge, Department of Data Sciences
8. Incharge, Department of Bioinformatics
9. Incharge, Department of Cyber Security
10. PS to Vice-Chancellor
11. PA to Registrar
12. Relevant File

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URL: http://www.uom.edu.pk

No. UOM/Acad/24/479 Dated: 05-Dec-2024

NOTIFICATION

It is notified for the information of all concerned that the Syndicate in its 57th meeting held on 02-Nov-2024 **[vide agenda item 24(3)]**, approved the following recommendations of the Academic Council made in its 22nd (Special) meeting held on 21-Oct-2024:

The courses reflected in the category of social sciences i.e., MGT 111, MGT 113, MGT 114 along with detailed courses be added to the General Education Courses of the University.

1. The course code of AI domain elective course at serial No. 50, titled as "AI Techniques for research and Future Technologies", be changed from "CS 365" to "AI 365".
2. The contact hours may be removed from the Scheme of Studies and only credit hours may be mentioned.
3. The committee also checked the HEC letter No. HEC/ACAD/UGEP/2024 dated 19-Sep-2024 regarding inclusion of the Course of Pakistan Studies in all undergraduate program as compulsory subjects. Therefore, the committee unanimously recommended that the Ideology and Constitution of Pakistan should be offered in all the scheme of studies in all BS programs as per the course contents attached with the HEC letter referred above.

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3. Deputy Registrar Admissions
4. Chairperson, Department of Artificial Intelligence
5. Chairperson, Department of Multimedia & Gamming
6. Incharge, Department of Data Sciences
7. Incharge, Department of Bioinformatics
8. Incharge, Department of Cyber Security
9. PS to Vice-Chancellor
10. PA to Registrar
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CURRICULUM FOR BS COMPUTER SCIENCE

(2024 onwards)



Department of Computer Science
University of Malakand

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Curriculum for BS Computer Science as per Revised NCEAC-2023 Criteria
(Effective from Fall 2024 and onwards)

Generic Structure for Computing Disciplines:

Category	Credit Hours	Courses
Computing Core (CC)	46	14
Domain Core (DC)	18	6
Domain Elective (DE)	21	7
Mathematics and Supporting Courses (MSC)	12	4
Elective Supporting Courses (ESC)	3	1
General Education Requirement (GER)	32	13
Total Cr. Hrs.	**132**	**45**

Mapping of *BS Computer Science* Program on the Generic Structure:

Computing Core (46/132): 14 Courses					
S.No.	**Code**	**Pre-Reqs**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	CS111		Programming Fundamentals	CC	4 (3-3)
2	CS121		Object Oriented Programming	CC	4 (3-3)
3	CS230		Data Structures	CC	4 (3-3)
4	CS240	CS120	Computer Organization and Assembly Language	CC	3 (2-3)
5	CS120		Digital Logic Design	CC	3 (2-3)
6	CS233		Operating Systems	CC	3 (2-3)
7	CS241	CS111	Database Systems	CC	4 (3-3)
8	CS471		Information Security	CC	3 (2-3)
9	CS232		Computer Networks	CC	3 (2-3)
10	CS242		Software Engineering	CC	3 (3-0)
11	CS362	CS230	Analysis of Algorithms	CC	3 (3-0)
12	CS350		Artificial Intelligence	CC	3 (2-3)
13	CS478		Final Year Project – I	CC	3 (0-9)
14	CS489		Final Year Project – II	CC	3 (0-9)

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Domain Core (18/132) 6 Courses					
S.No.	**Code**	**Pre-Reqs**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	CS353		Human Computer Interaction (HCI)	DC	3 (2-3)
2	CS473		Parallel and Distributed Computing	DC	3 (2-3)
3	CS352		Advanced Database Management Systems	DC	3 (2-3)
4	CS343		Theory of Automata	DC	3 (3-0)
5	CS367	CS343	Compiler Construction	DC	3 (2-3)
6	CS363	CS120	Computer Architecture	DC	3 (2-3)

Domain Elective (21/132): 7 Courses					
S.No.	**Code**	**Pre-Reqs**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	DS351	CS111	Python Programming	DE	3 (2-3)
2	CS479		Digital Image Processing	DE	3 (2-3)
3	CS365	CS111	Visual Programming in Java	DE	3 (2-3)
4	CS354		Web Design and Development	DE	3 (2-3)
5	CS355		Web Engineering	DE	3 (2-3)
6	MATH449		Numerical Analysis	DE	3 (3-0)
7	CS474		Cloud Computing	DE	3 (2-3)
8	CS483		Cyber Security	DE	3(2-3)
9	CS480	CS242	Software Testing and Quality Assurance	DE	3 (2-3)
10	CS475	CS121	Mobile App Development	DE	3 (2-3)
11	CS481		Computer Graphics	DE	3 (2-3)
12	CS477	CS242	Object Oriented Analysis and Design	DE	3(2-3)
13	CS368	CS121	Introduction to Modelling and Simulation	DE	3(2-3)
14	CS472	PHYS355	Internet of Things	DE	3(2-3)
Note: Core/ Elective Courses of 3 Cr.Hrs in other computing disciplines may be taken as elective courses					

Mathematics and Supporting Courses (MSC) (12/132) : 4 Courses					
S.No.	**Code**	**Pre-Reqs**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	MATH217		Multivariate Calculus	MSC	3 (3-0)
2	MATH114		Linear Algebra and Applications	MSC	3 (3-0)
3	STAT210		Probability and Statistics	MSC	3 (3-0)
4	ENG112		Technical and Business Writing	MSC	3 (3-0)

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Elective Supporting Courses (3/130): 1 Course					
S.No.	**Code**	**Pre-Reqs**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	MGT113		Social Science (Principles of Marketing)	ESC	3 (3-0)
2	MGT114		Social Science (Financial Accounting-I)	ESC	3 (3-0)

General Education Requirement as per HEC UG Education Policy (32/132): 13 Courses					
S.No.	**Code**	**Pre-Reqs**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	CS110		Introduction to Information and Communication Technologies	GER	3 (2-3)
2	ENG101		Functional English	GER	3 (3-0)
3	ENG102		Introduction to Expository Writing	GER	3 (3-0)
4	CS127		Quantitative Reasoning – 1 (Discrete Structures)	GER	3 (3-0)
5	MATH110		Quantitative Reasoning – 2 (Calculus and Analytical Geometry)	GER	3 (3-0)
6	ISL112		Islamic Studies/ Ethics (for Non-Muslims)	GER	2 (2-0)
7	PSC116		Pakistan Studies	GER	2 (2-0)
8	MGT111		Social Sciences (Principles of Management)	GER	2 (2-0)
9	PHYS355		Natural Sciences (Basic Electronics)	GER	3 (2-3)
10	CS311		Professional Practices	GER	2 (2-0)
11	SOC114		Civic and Community Engagements	GER	2 (2-0)
12	MGT215		Entrepreneurship	GER	2 (2-0)
13	ISL113		Seerah and its Contemporary Application سیرت رسول ﷺ اور اس کی عصری معنویت / Any course from arts and humanities in general category (for non-Muslims)	GER	2 (2-0)

Non-Credit (NC) Courses: 3 Courses					
S.No.	**Code**	**Pre-Reqs**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	MATH111		Mathematics I	NC	3 (3-0)
2	MATH122		Mathematics II	NC	3 (3-0)
3	ISL118		Teaching the Holy Quran with Translation	NC	2 (2-0)

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Scheme of Studies: BS Computer Science

Semester 1					
S.No.	**Code**	**Pre-Req.**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	ISL112		Islamic Studies/Ethics (for Non-Muslims)	GER	2 (2-0)
2	CS110		Introduction to Information and Communication Technologies	GER	2 (2-0)
3	CS110L		Introduction to Information and Communication Technologies Lab	GER	1 (0-3)
4	SOC114		Civic and Community Engagement	GER	2 (2-0)
5	PHYS355		Basic Electronics	GER	2 (2-0)
6	PHYS355L		Basic Electronics Lab	GER	1(0-3)
7	ENG101		Functional English	GER	3 (3-0)
8	CS111		Programming Fundamentals	CC	3 (3-0)
9	CS111L		Programming Fundamentals Lab	CC	1 (0-3)
10	MATH111		Mathematics I	NC	3 (3-0)
11	ISL118		Teaching the Holy Quran with Translation (Para 1,2,3)	NC	2 (2-0)
Total Cr. Hrs.					17

Semester 2					
S.No.	**Code**	**Pre-Req.**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	CS127		Discrete Structures	GER	3 (3-0)
2	MATH110		Calculus and Analytical Geometry	GER	3 (3-0)
3	MGT215		Entrepreneurship	GER	2 (2-0)
4	CS121	CS111	Object Oriented Programming	CC	3 (3-0)
5	CS121L		Object Oriented Programming Lab	CC	1(0-3)
6	ENG102		Introduction to Expository Writing	GER	3 (3-0)
7	CS120		Digital Logic Design	CC	2 (2-0)
8	CS120L		Digital Logic Design Lab	CC	1 (0-3)
9	MATH122		Mathematics II	NC	3 (3-0)
10	ISL124		Teaching the Holy Quran with Translation (Para 4,5,6, 7)	NC	2 (2-0)
Total Cr. Hrs.					18

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Semester 3					
S.No.	**Code**	**Pre-Req.**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	MATH217		Multivariate Calculus	MSC	3 (3-0)
2	CS230		Data Structures	CC	3 (3-0)
3	CS230		Data Structures Lab	CC	1 (0-3)
4	CS232		Computer Networks	CC	2 (2-0)
5	CS232L		Computer Networks Lab	CC	1 (0-3)
6	MGT111		Principles of Management	GER	2 (2-0)
7	CS233		Operating Systems	CC	2 (2-0)
8	CS233L		Operating Systems Lab	CC	1 (0-3)
9	ENG112		Technical and Business Writing	MSC	3 (3-0)
10	ISL226		Teaching the Holy Quran with Translation (Para 8, 9,10,11)	NC	2 (2-0)
Total Cr. Hrs.					18

Semester 4					
S.No.	**Code**	**Pre-Req.**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	PSC116		Pakistan Studies	GER	2 (2-0)
2	CS241	CS111	Database Systems	CC	3 (3-0)
3	CS241L		Database Systems Lab	CC	1 (0-3)
4	CS240		Computer Organization and Assembly Language	CC	2 (2-0)
5	CS240L		Computer Organization and Assembly Language	CC	1 (0-3)
6	CS242		Software Engineering	CC	3 (3-0)
7	CS243		Theory of Automata	DC	3 (3-0)
8	ISL113		Seerah and its Contemporary Application سیرت رسول ﷺ اور اس کی عصری معنویت / Any course from arts and humanities in general category (for non-Muslims)	GER	2 (2-0)
9	ISL240		Teaching the Holy Quran with Translation (Para 12, 13, 14, 15)	NC	2 (2-0)
Total Cr. Hrs.					17

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Semester 5					
S.No.	**Code**	**Pre-Req.**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	STAT210		Probability and Statistics	MSC	3 (3-0)
2	CS350		Artificial Intelligence	CC	2 (2-0)
3	CS350L		Artificial Intelligence Lab	CC	1 (0-3)
4	CS352		Advanced Database Management Systems	DC	2 (2-0)
5	CS352L		Advanced Database Management Systems Lab	DC	1 (0-3)
6	CS351		Professional Practices	GER	2 (2-0)
7			Domain Elective 1	DE	2 (2-0)
8			Domain Elective 1 Lab	DE	1 (0-3)
9	CS353	CS121	Human Computer Interaction (HCI)	DC	2 (2-0)
10	CS353L		Human Computer Interaction (HCI) Lab	DC	1 (0-3)
11	ISL337		Teaching the Holy Quran with Translation (Para 16, 17, 18, 19)	NC	2 (2-0)
Total Cr. Hrs.					17

Semester 6					
S.No.	**Code**	**Pre-Req.**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	MATH114		Linear Algebra and Applications	MSC	3 (3-0)
2	CS363	CS120	Computer Architecture	DC	3 (3-0)
3	CS367		Compiler Construction	DC	2 (2-0)
4	CS367L		Compiler Construction Lab	DC	1 (0-3)
5	CS362		Analysis of Algorithms	CC	3 (3-0)
6			Domain Elective 2	DE	2 (2-0)
7			Domain Elective 2 Lab	DE	1 (0-3)
8	ISL346		Teaching the Holy Quran with Translation (Para 20,21,22,23)	NC	2 (2-0)
Total Cr. Hrs.					15

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Semester 7					
S.No.	**Code**	**Pre-Req.**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	CS478		Final Year Project – I	CC	3 (0-9)
2			Domain Elective 3	DE	2 (2-0)
3			Domain Elective 3 Lab	DE	1 (0-3)
4			Domain Elective 4	DE	2 (2-0)
5			Domain Elective 4 Lab	DE	1 (0-3)
6	CS473		Parallel and Distributed Computing	DC	2 (2-0)
7	CS473L		Parallel and Distributed Computing Lab	DC	1 (0-3)
8	CS471		Information Security	CC	3 (3-0)
9	ISL447		Teaching the Holy Quran with Translation (Para 24,25,26,27)	NC	2 (2-0)
Total Cr. Hrs.					15

Semester 8					
S.No.	**Code**	**Pre-Req.**	**Course Title**	**Category**	**Cr. Hrs. (Contact Hrs.)**
1	CS489		Final Year Project – II	CC	3 (0-9)
2			Domain Elective 5	DE	2 (2-0)
			Domain Elective 5 Lab	DE	1 (0-3)
3	MATH449		Domain Elective 6 (Numerical Analysis)	DE	3 (3-0)
4			Domain Elective 7	DE	2 (2-0)
			Domain Elective 7 Lab	DE	1 (0-3)
5	MGT113		Principles of Marketing	ESC	3 (3-0)
5	ISL455		Teaching the Holy Quran with Translation (Para 28,29,30)	NC	2 (2-0)
Total Cr. Hrs.					15
Program Total Cr. Hrs.					**132**

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Minor in Computer Science for other Disciplines:

Four Courses of Minimum 12 Cr. Hrs.

S. No.	Code	Pre-Req.	Course Title	Category	Cr. Hrs. (Contact Hrs.)
1	CS111		Programming Fundamentals	Core	3 (3-0)
2	CS111L		Programming Fundamentals Lab	Core	1 (0-3)
5	CS121	CS111	Object Oriented Programming	Core	3 (3-0)
6	CS121L		Object Oriented Programming Lab	Core	1(0-3)
3	CS241		Database Systems	Core	3 (3-0)
4	CS241L		Database Systems Lab	Core	1 (0-3)
7	CS365	CS111	Visual Programming in Java	Elective	2 (2-0)
8	CS365L		Visual Programming in Java Lab	Elective	1 (0-3)
9	DS351	CS111	Python Programming	Elective	2 (2-0)
10	DS351L		Python Programming Lab	Elective	1 (0-3)
11	CS360	CS111	Web Design and Development	Elective	2 (2-0)
12	CS360L		Web Design and Development Lab	Elective	1 (0-3)
Total Cr. Hrs.					12 to 15

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Program Educational Objectives (PEOs):

PEO-1: Engage in successful professional practice in their chosen field and demonstrate professional advancement through significant technical achievements by staying current with emerging trends and technologies in the field through lifelong learning.

PEO-2: Demonstrate professional attitudes, ethics, and ability to work effectively as a team member and/or leader in an ever-changing professional environment;

PEO-3: Pursue advanced degrees in computing, sciences, engineering, business, or other professionally related fields, and will engage in life-long learning and professional development.

Program Learning Outcomes (PLOs)

PLO-1 Academic Education
Completion of an accredited program (BS CS) of study designed to prepare graduates as computing professionals

PLO-2 Knowledge for Solving Computing Problems
Apply knowledge of computing fundamentals, knowledge of computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements

PLO-3 Problem Analysis
Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines

PLO- 4 Design/ Development of Solutions
Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

PLO-5 Modern Tool Usage
Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations

PLO-6 Individual and Team Work
Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings

PLO-7 Communication
Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions

PLO-8 Computing Professionalism and Society
Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice

PLO-9 Ethics
Understand and commit to professional ethics, responsibilities, and norms of professional computing practice

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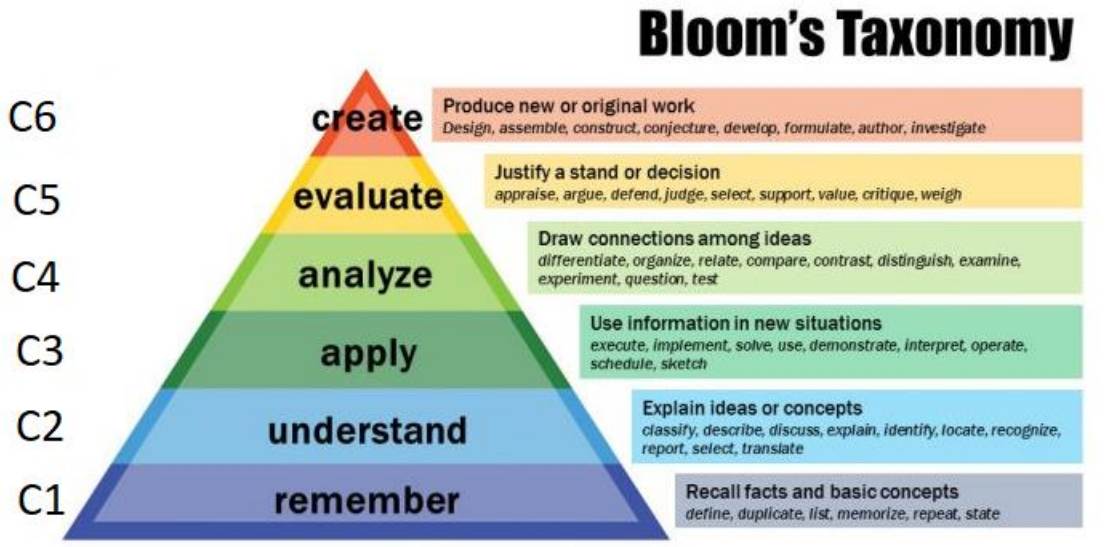
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PLO-10 Life-long Learning
Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional

Program Learning Outcomes (PLOs) and their mappings to PEOs:

Program Educational Objectives (PEOs)				
PLOs	**Program Learning Outcomes (PLOs)**	**PEO-1**	**PEO-2**	**PEO-3**
PLO-1	Academic Education	✓		
PLO-2	Knowledge for Solving Computing Problems	✓		
PLO-3	Problem Analysis	✓		
PLO-4	Design/Development of Solutions	✓		
PLO-5	Modern Tool Usage	✓		
PLO-6	Individual and Team Work		✓	
PLO-7	Communication		✓	
PLO-8	Computing Professionalism and Society		✓	✓
PLO-9	Ethics		✓	
PLO-10	Lifelong Learning	✓		✓

Cognitive Levels: Bloom's Taxonomy



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Computing Core Subjects

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Course Name: Programming Fundamentals

Credit Hours: 4 (3-3)

Course Introduction:

This course provides fundamental concepts of programming to freshmen. The courses is pre-requisite to many other courses, therefore, students are strongly advised to cover all contents and try to achieve CLOs to the maximum possible level. The course may be taught as programing language independent. The teacher can choose any programing language for the practical/lab purpose but that must be latest and market oriented.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand basic problem solving steps and logic constructs	C2 (Understand)	2
CLO-2	Apply basic programing concepts	C3 (Apply)	5
CLO-3	Design and implement algorithms to solve real world problems	C3 (Solve)	4

Course Outline:

Introduction to problem solving, a brief review of Von-Neumann architecture, introduction to programming, role of compiler and linker, introduction to algorithms, basic data types and variables, input/output constructs, arithmetic, comparison and logical operators, conditional statements and execution flow for conditional statements, repetitive statements and execution flow for repetitive statements, lists and their memory organization, multi- dimensional lists, introduction to modular programming, function definition and calling, stack rolling and unrolling, string and string operations, pointers/references, static and dynamic memory allocation, File I/O operations.

Reference Materials (or use any other standard and latest books):

- Starting out with Programming Logic and Degins, Latest Edition by Tony Gaddis,
- The C Programming Language, Latest Edition Edition by Brian W. Kernighan, Dennis M. Ritchie
- Object Oriented Programming in C++ , Latest Edition by Robert Lafore
- C How to Program, Latest Edition by Paul Deitel and Harvey Deitel
- Problem Solving and Program Design in C++, Latest Edition by Jeri R. Hanly and Elliot B. Koffman

Course Name: Programming Fundamentals	
Week #	**Week wise distribution**
1	Overview of Computer Programming, Principles of Structured and Modular Programming, Overview of Structured Programming Languages, Algorithms and Problem Solving, Program Development: Analyzing Problem
2	Designing Algorithm/Solution, Testing Designed Solution, Translating
3	Algorithms into Programs
4	Fundamental Programming Constructs, Data Types. Basics of Input and Output
5	Selection and Decision If, If -Else, Nested If- Else, Switch Statement and Condition Operator)
6	Repetition While and For Loop

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7	Do -While Loops
8	Break Statement, Continue Statement
9	Control Structures
10	Functions
11	Arrays
12	Pointers
13	Records
14	Files Input-Output
15	Testing and Debugging
16	Wrap up work and presentations

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Course Name: Object Oriented Programming
Credit Hours: 4 (3-3)

Course Introduction:
The course aims to focus on object-oriented concepts, analysis and software development. The basic concept of OOP is covered in this course.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand principles of object oriented paradigm	C2 (Understand)	2
CLO-2	Identify objects and their relationships to design object oriented solution	C2 (Identify), C6(Design)	3, 4
CLO-3	To implement an object oriented solution	C3 (Apply)	5

Course Outline:
Introduction to object oriented design, history and advantages of object oriented design, introduction to object oriented programming concepts, classes, objects, data encapsulation, constructors, destructors, access modifiers, const vs non-const functions, static data members and functions, function overloading, operator overloading, identification of classes and their relationships, composition, aggregation, inheritance, multiple inheritance, polymorphism, abstract classes and interfaces, generic programming concepts, function and class templates, standard template library, object streams, data and object serialization using object streams, exception handling.

Reference Materials: (or use any other standard and latest books)
- Java: How to Program, Latest Edition by Paul Deitel
- Beginning Java 2, Latest Edition by Ivor Horton
- An Introduction to Object Oriented Programming with Java, Latest Edition by C. Thomas Wu
- Starting Out with C++ from Control Structures to Objects, Latest Edition, Tony Gaddis
- C++ How to Program, Latest Edition, Deitel and Deitel.
- Object Oriented Programming in C++, Latest Edition by Robert Lafore.

Course Name: Object Oriented Programming	
Week #	**Week wise distribution**
1	Evolution of Object Oriented Programming (OOP), Object Oriented concepts and principles
2	Problem solving in Object Oriented paradigm
3	OOP design process
4	Classes, functions/methods
5	Objects and encapsulation
6	Constructors and destructors
7	Operator and function/method overloading
8	Association, aggregation, composition, generalization,
9	Inheritance and its types
10	Derived classes
11	Function/method overriding
12	Abstract and concrete classes
13	Virtual functions

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14	Polymorphism
15	Exception handling.
16	Wrap up work and presentations

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Course Name: Data Structures
Credit Hours: 4 (3-3)

Course Introduction:
The course is designed to teach students data structures and algorithms, which allow them to write programs to efficiently manipulate, store, and retrieve data. Students will learn the concepts of time and space complexity of algorithms and computer programs.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	To understand various data structures and their algorithms	C2 (Understand)	2
CLO-2	Analyze various algorithmic techniques to solve complex computational problems efficiently.	C4 (Analyze)	3
CLO-3	Apply the knowledge of data structure to other application domains	C3 (Apply)	5

Course Outline:
Data Structure, Types, Algorithm, Algorithm Notation, Data Structure Operations, Complexity of an Algorithms, Big Oh notation, Arrays, Types of an Array and their Representation in Memory, Searching an array, Binary search for sorted arrays, Linked list and its various types, Sorted linked list, Stacks (linked lists and array implementations), Recursion and analyzing recursive algorithms, Queue, Double ended queue, Priority queues (linked and array implementations of queues), Trees and tree traversals, Binary search trees, Heaps, Balanced trees, Graphs, Breadth-first and depth-first traversal, Topological order, Shortest path, Graph Representations, Sorting algorithms (selection, insertion, merge, quick, bubble, heap, shell, radix), Hashing and indexing, Open addressing and chaining, Memory management and garbage collection.

Reference Materials: (or use any other standard and latest books)
- Data Structures and Algorithm Analysis in Java by Mark A. Weiss, Latest Edition
- Data Structures and Abstractions with Java by Frank M. Carrano and Timothy M. Henry, Latest Edition
- Data Structures and Algorithms in C++ by Adam Drozdek, Latest Edition
- Data Structures and Algorithms Shaums Outline Series by Lipchautz, Latest Edition
- Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss, Latest Edition

Course Name: Data Structures	
Week #	**Week wise distribution**
1.	Introduction to Data Structures and Algorithms, Basic Terminology, Algorithm Writing Notation, Data Structure operations, Complexity Analysis.
2.	Arrays (Linear, Two Dimensional and Multi-Dimensional, Memory Representation of

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	Array.
3.	Operations on Array (Traversing, Insertion, Deletion, Searching, Sorting)
4.	Sorting: (Bubble Sort. Insertion Sort, Selection Sort, Shell Sort, Heap Sort
5.	Quick Sort, Merge Sort, Radix Sort)
6.	Linked Lists: Singly Linked Lists, Doubly Linked Lists, Circular Linked List, Implantation of Linked List
7.	Stacks, Queues, Priority Queue and their implementation.
8.	Recursion: Function call and Recursion Non-tail Recursion, Indirect Recursion, Nested Recursion, Backtracking, Implementation,
9.	Trees: Introduction, Representation in memory, Types, Implementation
10.	Binary Heap, Binary Search Tree
11.	Tree Traversal, Insertion, Deletion, and Balancing a Tree
12.	Heap. B-Tree, B + Tree, Spanning Tree, Splay Trees, Huffman Coding
13.	Graphs: Representation, Graph Traversal Algorithms, Shortest Path Algorithm.
14.	Isomorphic Graphs. Hashing techniques,
15.	Hashing and Hashing Techniques, Memory Management and Garbage Collection
16.	Wrap up work and presentations

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Course Name: Computer Organization and Assembly Language
Credit Hours: 3 (2-3)

Course Introduction:
The main objective of this course is to introduce the organization of computer systems and usage of assembly language for optimization and control. Emphasis should be given to expose the low-level logic employed for problem solving while using assembly language as a tool. At the end of the course the students should be capable of writing moderately complex Assembly language subroutines and interfacing them to any high-level language.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Acquire the basic knowledge of computer organization, computer architecture and assembly language	C2 (Understand)	2
CLO-2	Usage of Assembly language to access basic computer hardware	C3 (Apply)	5
CLO-3	To write Assembly language programs for problems related to computer organization	C6 (Create)	4,5

Course Outline:
Organization and design of Basic Computer, Microprocessor Bus Structure: Address, Data and Control, Segmented Memory Model, Objectives and Perspectives of Assembly Language, Data Declaration, Addressing Modes, Direct Addressing, Indirect Addressing, Branching, Unconditional jumps and relative addressing, Bit Manipulations, Multiplication in Assembly Language, Bitwise Logical Operations and Masking Operations, Subroutines, Stack and its Operations, Parameter Passing Through Stack and Local Variables, Display Memory, Display Memory and Number Printing in Assembly, Screen Location Calculation, Clearing the Screen and String Printing, Screen Scrolling and string Comparison, Interrupts Handling, BIOS and DOS Interrupts, Hardware Interrupts, Ports, Interrupt Chaining and Terminate and Stay Resident, Programmable Interval Timer, BIOS Video Services, DOS Video Services 2, 32bit Programming, Interfacing with High level languages, Processor architecture: ARM Architecture

Reference Materials:
- Computer System Architecture, M. Morris Mano, Latest Edition,
- Assembly Language Programming for Intel- Computer, Latest Edition
- Computer Systems: A Programmer's Perspective, 3/E (CS:APP3e), Randal E. Bryant and David R.O' Hallaron, Carnegie Mellon University
- Robert Britton, MIPS Assembly Language Programming, Latest Edition,

Course Name: Computer Organization and Assembly Language	
Week #	**Week wise distribution**
1	Organization and design of Basic Computer, Microprocessor Bus Structure: Address, Data and Control, Segmented Memory Model
2	Objectives and Perspectives of Assembly Language, Data Declaration and Addressing Modes, Direct Addressing
3	Indirect Addressing
4	Branching, Unconditional jumps and relative addressing
5	relative addressing, Bit Manipulations
6	Multiplication in Assembly Language, Bitwise Logical Operations and Masking Operations
7	Subroutines, Stack and its Operations, Parameter Passing Through Stack and Local

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	Variables
8	Display Memory, Display Memory and Number Printing in Assembly
9	Screen Location Calculation, Clearing the Screen and String Printing, Screen Scrolling and string Comparison
10	Interrupts Handling, BIOS and DOS Interrupts
11	Hardware Interrupts, Ports, Interrupt Chaining and Terminate and Stay Resident, Programmable Interval Timer
12	BIOS Video Services
13	DOS Video Services 2
14	32bit Programming, Interfacing with High level languages
15	Processor architecture: ARM Architecture
16	Wrap up work and presentations

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Course Name: Digital Logic Design
Credit Hours: 3 (2-3)

Course Introduction:
The course introduces the Boolean algebra, digital logic, gates and digital circuits. Further, it focuses on the design and analysis of combinational and sequential circuits. It also serves to familiarize the student with the logic design of basic computer hardware components.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Acquire knowledge related to the concepts, tools and techniques for the design of digital electronic circuits	C2 (Understand)	2
CLO-2	To design and create logic circuits for practical implementations.	C6 (Create)	4
CLO-3	Demonstrate the skills to analyze both combinational and sequential circuits using a variety of Techniques	C4 (Analyze)	3
CLO-4	Apply the acquired knowledge to simulate and implement small scale digital circuits	C3 (Apply)	5

Course Outline:
Number Systems, Logic gates, Boolean algebra, Combination logic circuits and designs, Simplification Methods (K-Map, Quinn Mc-Cluskey method), Flip Flops and Latches, Asynchronous and Synchronous circuits, Counters, Shift Registers, Counters, Triggered devices and its types. Mealy machines and Moore machines. Binary Arithmetic and Arithmetic circuits, Memory elements, State Machines. Introduction to Programmable Logic Devices (CPLD, FPGA) Lab Assignments using tools such as Verilog HDL/VHDL, MultiSim.

Reference Materials:
- Digital Fundamentals by Floyd, Latest Edition.
- Fundamental of Digital Logic with Verilog Design, Stephen Brown, Latest Edition.

Course Name: Digital Logic Design	
Week #	**Week wise distribution**
1	Number Systems, Logic Gates, Boolean Algebra
2	Combination logic circuits and designs
3	Algebra based simplification
4	K-Maps based Simplification Methods
5	Quinne Mc-Cluskey Method
6	Flip Flops and Latches
7	Asynchronous and Synchronous circuits
8	Counters
9	Shift Registers
10	Shift Registers Counters
11	Triggered devices and its types
12	Binary Arithmetic and Arithmetic Circuits, Memory Elements
13	State Machines
14	Introduction Programmable Logic Devices CPLD, FPGA)
15	Lab Assignments using tools such as Verilog HDL/VHDL, MultiSim, etc.
16	Wrap up work and presentations

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Course Name: Operating Systems
Credit Hours: 3 (2-3)

Course Introduction:
To prepare students to gain a general understanding of the principles and concepts governing the functions of operating systems (OS) and acquaint students with the layered approach that makes design, implementation and operation of the complex OS.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand the characteristics of different structures of the Operating Systems and identify their core functions	C2 (Understand)	2
CLO-2	Analyze and evaluate the algorithms of the core functions of the Operating Systems and explain the major performance issues with regard to the core functions	C5 (Evaluate)	2, 3
CLO-3	Demonstrate the knowledge in applying system software	C3 (Demonstrate)	4, 5

Course Outline:
Definition and purpose of an operating system, Types of operating systems: Batch OS, Time-sharing OS, Distributed OS, Real-time OS, Embedded OS, Network OS, Mobile OS, Components of an operating system, Role of an operating system in a computer system, system calls, process concept and scheduling, inter-process communication, multithreaded programming, multithreading models, threading issues, process scheduling algorithms, thread scheduling, multiple-processor scheduling, synchronization, critical section, synchronization hardware, synchronization problems, deadlocks, detecting and recovering from deadlocks, memory management, swapping, contiguous memory allocation, segmentation and paging, virtual memory management, demand paging, thrashing, memory-mapped files, file systems, file concept, directory and disk structure, directory implementation, free space management, disk structure and scheduling, swap space management, system protection, virtual machines, operating system security.

Reference Materials:
- Operating Systems Concepts, Latest Edition by Abraham Silberschatz
- Modern Operating Systems, Latest Edition by Andrew S. Tanenbaum
- Operating Systems, Internals and Design Principles, Latest Edition by William Stallings

Course Name: Operating Systems	
Week #	**Week wise distribution**
1	Definition and purpose of an operating system, Types of operating systems: Batch OS, Time-sharing OS, Distributed OS, Real-time OS, Embedded OS, Network OS, Mobile OS,
2	Operating System: Services, Structure, User Interface
3	Virtual Machines concept, System Boot, System Calls
4	Types of System Calls, Processes: Concept, Scheduling,
5	Operations on Processes, Inter-process Communication. Threading: Multithreading Models,
6	Thread Libraries, Threading Issues, processor scheduling:
7	Deadlock prevention, avoidance, and recovery
8	Main-memory management; virtual memory management swapping,
9	Paging, segmentation and page replacement algorithms
10	Disks management and other input/output devices
11	File-system structure and implementation;

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12	File-system structure and implementation;
13	Protection and Security
14	Case studies: Linux/Windows Operating Systems.
15	Case studies: Linux/Windows Operating Systems.
16	Wrap up work and presentations

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Course Name: Database Systems
Credit Hours: 4 (3-3)

Course Introduction:
The course aims to introduce basic database concepts, different data models, data storage and retrieval techniques, and database design techniques. The course primarily focuses on relational data model and Data Base Management Systems (DBMS) concepts.

CLO No	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand fundamental database concepts.	C2 (Understand)	2
CLO-2	Design conceptual, logical and physical database schemas using different data models.	C6 (Design)	4
CLO-3	Use Structured Query Language (SQL) for database definition and manipulation in a DBMS	C3 (Apply)	4, 5

Course Outline:
Basic database concepts, Database approach vs. file based system, database architecture, three level schema architecture, data independence, relational data model, entity relationship model, entity sets, attributes, relationship, entity-relationship diagrams attributes, schemas, tuples, domains, relation instances, keys of relations, integrity constraints, relational algebra, selection, projection, Cartesian product, types of joins, normalization, functional dependencies, normal forms, Structured Query Language (SQL), Joins and sub-queries in SQL, Grouping and aggregation in SQL, indexes.

Reference Materials:
- Modern Database Management, Jeffrey A. Hoffer et al., Latest Edition
- Database Systems: A Practical Approach to Design, Implementation, and Management by Thomas Connolly and Carolyn Begg, Latest Edition.
- Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom, Latest Edition
- Database System Concepts by Avi Silberschatz, Henry F. Korth and S. udarshan Latest Edition.
- Database Management Systems by Raghu Ramakrishnan, Johannes Gehrke, Latest Edition

Course Name: Database Systems	
Week #	**Week wise distribution**
1	Basic Database Concepts, Database Architecture, DB Design Life Cycle
2	Schema Architecture, Conceptual Schema
3	Logical and Physical Database Modeling and Design
4	Entity Relationship Diagram ERD
5	Enhanced ERD
6	Relational Data Model
7	Mapping ERD to Relations
8	Functional Dependencies
9	MID TERM EXAMS
10	Normalization
11	Physical Database Design

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12	Relational Algebra
13	Structured Query Language SQL)
14	SQL Continued
15	Transaction Processing
16	Query Optimization Concepts

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Course Name: Information Security
Credit Hours: 3 (2-3)

Course Introduction:
This course provides a broad overview of the threats to the security of information systems, the responsibilities and basic tools for information security, and the levels of training and expertise needed in organizations to reach and maintain a state of acceptable security. It covers concepts and applications of system and data security. Areas of particular focus include secure network design, implementation and transition issues, and techniques for responding to security breaches.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand key concepts of information security, legal, ethical, and professional issues in information security	C2 (Understand)	2
CLO-2	Apply various security and risk management tools for achieving information security and privacy	C3 (Apply)	5
CLO-3	Identify appropriate techniques to tackle and solve problems in the discipline of information security	C4 (Identify)	2, 3, 4

Course Outline:
Information security foundations, security design principles; security mechanisms, symmetric and asymmetric cryptography, encryption, hash functions, digital signatures, key management, authentication and access control; software security, vulnerabilities and protections, malware, database security; network security, firewalls, intrusion detection; security policies, policy formation and enforcement, risk assessment, cybercrime, law and ethics in information security, privacy and anonymity of data.

Reference Materials: (or use any other standard and latest books)
- Computer Security: Principles and Practice, Latest edition by William Stallings
- Principles of Information Security, Latest edition by M. Whitman and H. Mattord
- Computer Security, Latest edition by Dieter Gollmann
- Computer Security Fundamentals, Latest edition by William Easttom
- Official (ISC) 2 Guide to the CISSP CBK, Latest edition

Course Name: Information Security	
Week #	**Week wise distribution**
1	Basic notions of confidentiality, integrity, availability, authentication models
2	protection models, security kernels, Encryption, Hashing and Digital Signatures
3	audit; intrusion detection and response, database security, host-based
4	network-based security
5	issues operational security issues
6	physical security issues
7	personnel security
8	policy formation and enforcement
9	access controls; information flow
10	legal and social issues identification and authentication in local and distributed systems
11	Continue…
12	Continue…

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13	Continue…
14	classification and trust modeling
15	risk assessment
16	Wrap up work and presentations

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Course Name: Computer Networks
Credit Hours: 3 (2-3)

Course Introduction:
This course introduces the basic concept of computer networks, Network layers, Network models (OSI, TCP/IP) and protocol standards.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand the concepts of technologies of computer networks and services and functions of each layer in the Internet protocol stack	C2 (Understand)	2
CLO-2	Identify various internetworking devices and protocols and their functions in a networking	C2 (Identify)	2
CLO-3	Analyze working and performance of key technologies, algorithms and protocols	C4 (Analyze)	3
CLO-4	Build Computer Network on various Topologies	C6 (Design)	4

Course Outline:
Introduction and protocols architecture, basic concepts of networking, network topologies, layered architecture, physical layer functionality, data link layer functionality, multiple access techniques, circuit switching and packet switching, LAN technologies, wireless networks, MAC addressing, networking devices, network layer protocols, IPv4 and IPv6, IP addressing, sub-netting, CIDR, routing protocols, transport layer protocols, ports and sockets, connection establishment, flow and congestion control, application layer protocols, latest trends in computer networks.

Reference Materials: (or use any other standard and latest books)
- Computer Networking: A Top-Down Approach Featuring the Internet, by James F. Kurose and Keith W. Ross. (Latest Editions)
- Computer Networks, Edition by Andrew S. Tanenbaum. (Latest Editions)
- Data and Computer Communications, Edition by William Stallings. (Latest Editions)
- Data Communication and Computer Networks, Edition by Behrouz A. Forouzan. (Latest Editions)

Course Name: Computer Networks	
Week #	**Week wise distribution**
1	Data communication concepts
2	Analogue and digital Transmission, Noise
3	Media
4	Encoding
5	Asynchronous and Synchronous transmission.
6	Wireless LANs
7	Lab exercises using tools such as Wireshark, OpNet, Packet tracer etc.
8	Network system architectures OSI, TCP/I P
9	Error control
10	Flow control
11	Data Link Protocols

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12	Bridging. Local area networks (LAN)
13	MAC Layer protocols, Multiplexing, Switching
14	IP Networks, Inter-networking, Routing, Transport layer protocols TCP
15	UDP and SCTP, Application layer protocols
16	Wrap up work and presentations

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Course Name: Software Engineering
Credit Hours: 3 (3-0)

Course Introduction: This course introduces the basic concepts and principles of Software Engineering. The course also covers various software paradigms for design and development of software.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand various software engineering processes and activities in software development life cycle	Cs (Understand)	2
CLO-2	Apply software modeling techniques to develop a software system	C3 (Apply)	3, 6
CLO-3	Apply software quality assurance and testing principles to evaluate software systems	C4 (Analyze), C5 (Evaluate)	4, 5
CLO-4	Apply key principles and common methods to analyze and maintain software systems	C3 (Apply), C4 (Analyze)	3, 4

Course Outline:
Definition and importance of software engineering, overview of software engineering, software crises, nature and types of software, software application domains, software development methodologies, functional and non-functional requirements, overview of the software development lifecycle (SDLC), software process models, understanding software requirements, requirements engineering, gathering and analyzing requirements, writing effective requirements specifications, use case diagrams and user stories, software engineering practice, software design concepts, patterns and best practices, high-level vs detailed design, software architectural design, software testing techniques and quality assurance, types of testing (unit, integration, system, acceptance etc.), object-oriented design, principles of software design, design patterns, uml diagrams (class, sequence, activity), software maintenance and evolution, types of software maintenance, refactoring techniques, managing legacy systems, team collaboration and communication, effective communication in software teams, collaboration tools (slack, microsoft teams), conflict resolution, ethics and professional practice, ethical issues in software engineering, professional responsibilities.

Reference Materials: (or use any other standard and latest books)

- Software Engineering, Sommerville I., Latest Edition, Pearson Inc.
- Software Engineering, A Practitioner's Approach, Pressman R. S.and Maxim B. R., Latest Edition, McGraw-Hill.

Week #	Week wise distribution
1	Definition of software engineering, importance of software engineering in modern development, Overview of the discipline and its role in producing high-quality software
2	Historical perspective: The software crisis, Nature and types of software The challenges faced in early software development
3	Various application domains: embedded systems, real-time systems, web applications.
4	Understanding functional requirements (what the system should do), Non-functional requirements (performance, usability, security, etc.), Importance of both in system development
5	Phases of the SDLC: requirements, design, implementation, testing, maintenance, How SDLC governs the software development process
6	Introduction to process models: Waterfall, Agile, V-Model, Incremental, etc. choosing the right model for the project context
7	Why requirements are critical to software success, Types of requirements (business, user, system), Defining clear, actionable requirements

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8	Techniques for gathering and analyzing requirements, Writing effective requirements specifications, Best practices for refining and documenting requirements
9	Introduction to use case diagrams and their role in system design, Writing user stories for Agile development, Practical examples of creating both
10	Key practices in software engineering for delivering quality projects, Principles of software design (modularity, abstraction, encapsulation), Balancing design considerations and constraints
11	Difference between high-level and detailed design, Introduction to software architectural patterns (client-server, layered architecture, etc.), Creating effective software architectures
12	Importance of testing in the software lifecycle, Different types of testing: unit, integration, system, acceptance, The role of quality assurance in ensuring software quality
13	Principles of object-oriented design: encapsulation, inheritance, polymorphism, Introduction to UML diagrams: class, sequence, activity diagrams, How to model software systems visually using UML
14	Types of software maintenance: corrective, adaptive, perfective, preventive, Managing legacy systems and keeping software up to date, Refactoring techniques to improve code quality over time
15	Effective communication in software teams, Collaboration tools (Slack, Microsoft Teams), Conflict resolution strategies for software teams, Ethical issues in software development, The professional responsibilities of a software engineer
16	Wrap up work and presentations

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Course Name: Analysis of Algorithms
Credit Hours: 3 (3-0)

Course Introduction:
The course introduces the basic notions of the design of algorithms and the underlying data structures and several measures of complexity.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand and explain the behavior of algorithms under different conditions and assumptions.	C2 (Understand)	2
CLO-2	Analyze and determine the time and space complexity of algorithms.	C4 (Analyze)	3
CLO-3	Apply algorithmic strategies to solve computational problems.	C3 (Apply)	4,5

Course Outline:
Introduction; role of algorithms in computing, Analysis on nature of input and size of input Asymptotic notations; Big-O, Big Ω, Big Θ, little-o, little-ω, Sorting Algorithm analysis, loop invariants, Recursion and recurrence relations; Algorithm Design Techniques, Brute Force Approach, Divide-and-conquer approach; Merge, Quick Sort, Greedy approach; Dynamic programming; Elements of Dynamic Programming, Search trees; Heaps; Hashing; Graph algorithms, shortest paths, sparse graphs, String matching; Introduction to complexity classes.

Reference Materials:
- Introduction to Algorithms by Thomas H. Corman, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein. (Latest editions)
- Algorithm Design, Jon Kleinberg, Eva Tardos. (Latest editions)
- Algorithms, Robert Sedgewick, Kevin Wayne. (Latest editions)

Course Name: Analysis of Algorithms	
Week #	**Week wise distribution**
1.	Introduction to analysis of algorithms and its advantages with examples, How algorithms are analyzed with simple examples
2.	Analysis of Insertion sort, bubble sort, selection sort
3.	Analysis of searching algorithms (Linear search, Binary search)
4.	Order of growth, Asymptotic notations, Formal definitions of asymptotic notations and their proofs, examples
5.	Recursion and recurrence relations; Iteration method
6.	Recursion and recurrence relations; Recursion tree method
7.	Recursion and recurrence relations; Master Theorem method
8.	Divide-and-conquer approach;
9.	Sorting; Search trees;
10.	Heaps;

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11.	Hashing;
12.	Dynamic programming; Graph algorithms;
13.	Greedy approach;
14.	Shortest paths; Network flow; Disjoint Sets;
15.	Polynomial and matrix calculations;
16.	String matching; NP complete problems; Approximation algorithms.

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Course Name: Artificial Intelligence
Credit Hours: 3 (2-3)

Course Introduction:
This course focuses on the foundations of AI and its basic techniques like Symbolic manipulations, Pattern Matching, Machine Learning, Computer Vision, Knowledge Representation, Decision Making and Appreciating the differences between Knowledge and Data.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand AI Foundations and Techniques	C2(Understand)	2
CLO-2	Analyze various AI algorithms for problem solving	C4 (Analyze)	3
CLO-3	Implement artificial intelligence techniques and case studies using Python programming	C3 (Design)	4, 5

Course Outline:
Introduction to Artificial Intelligence and its applications towards Knowledge Based Systems; Introduction to Reasoning and Knowledge Representation, Problem Solving by Searching (Informed searching, Uninformed searching, Heuristics, Local searching, Min- max algorithm, Alpha beta pruning, Game-playing); Case Studies: General Problem Solver, Eliza, Student, Macsyma; Learning from examples; ANN and Natural Language Processing; Recent trends in AI and applications of AI algorithms. Python programming language will be used to explore and illustrate various issues and techniques in Artificial Intelligence.

Reference Materials: (or use any other standard and latest books)

- Russell, S. and Norvig, P. "Artificial Intelligence. A Modern Approach", Latest Edition, Prentice Hall, Inc.
- Severance, C.R. "Python for everybody: Exploring data using Python 3." CreateSpace Independent Publ Platform. Latest Edition
- Miller, B.N., Ranum, D.L. and Anderson, J. "Python programming in context." Jones and Bartlett Pub. Latest Edition
- Joshi, P. "Artificial intelligence with python." Packt Publishing Ltd. Latest Edition

Course Name: Artificial Intelligence	
Week #	**Week wise distribution**
1	Artificial Intelligence: Introduction, AI Paradigms and Hypothesis
2	Intelligent Agents. Difference between Cybernetic Intelligence and Artificial Intelligence
3	Objectives and Scope of Weak AI and Strong AI, Problem-solving
4	Solving Problems by Searching, Informed Search and Exploration
5	Constraint Satisfaction Problems, Adversarial Search. Knowledge and reasoning
6	Logical Agents, First-Order Logic, Inference in First-Order Logic, Knowledge Representation
7	Planning and Acting in the Real World. Uncertain knowledge and reasoning, Uncertainty,
8	Probabilistic Reasoning, Probabilistic Reasoning over Time, Making Simple Decisions
9	Making Complex Decisions. Learning: Learning from Observations, Knowledge in Learning
10	Learning Methods, Reinforcement Learning. Communicating, perceiving, and acting,

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	Communication
11	Probabilistic Language Processing, Perception and Robotics
12	Introduction to Expert Systems and Applications
13	Artificial General Intelligence
14	Issues in Safe AI
15	Introduction to Cognitive and Conscious Systems
16	Wrap up work and presentations

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Domain Core Subjects

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Course Name: Human Computer Interaction (HCI)

Credit Hours: 3 (2-3)

Course Introduction:

This course covers understanding the area of Human Computer Interaction by introducing various methods and techniques that could be used to design and evaluate interactive systems.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand the basics of human and computational abilities and limitations	C2 (Understand)	2
CLO-2	Understand basic theories, tools and techniques in HCI.	C2 (Understand)	2,5
CLO-3	Understand the fundamental aspects of designing and evaluating interfaces.	C2 (Understand)	3, 4
CLO-4	Apply appropriate HCI techniques to design systems that are usable by people.	C3 (Apply)	4

Course Outline:

The human and the computer and their interaction, Human psychology and ergonomics, Interaction Paradigms, Interaction design basics, HCI in the software process, Design rules, Implementation support, Evaluation techniques, Universal design, User support, Cognitive models, Socio-organizational issues and stakeholder requirements, Communication and collaboration models, Task analysis, Dialog notations and design , Models of the system, Modeling rich interaction, Groupware, Ubiquitous computing and augmented realities

Reference Materials:

- Human-Computer Interaction by Alan Dix, Janet E. Finlay, Gregory D. Abowd, Russell Beale, Prentice Hall; Latest Edition
- Human-Computer Interaction: Concepts And Design by J. Preece, Y. Rogers, H. Sharp, D. Benyon, S. Holland, T. Carey, Addison Wesley; Latest Edition
- Human-Computer Interaction Handbook: Fundamentals, Evolving Technologies, and Emerging Applications by Julie A. Jacko, CRC Press; Latest Edition
- Interaction Design: Beyond Human - Computer Interaction by Yvonne Rogers, Helen Sharp, and Jenny Preece, Wiley; Latest Edition
- Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Rules by Jeff Johnson, Morgan Kaufmann; Latest Edition.

	Course Name: HCI
Week #	**Week wise distribution**
1.	The human perception capabilities and limitations, Human memory, cognition, reasoning and problem solving
2.	The Computer (Input devices, display devices , paper output, memory and processing)
3.	Models of interaction, Framework and HCI, Interaction styles, Elements of WIMP

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4.	Interaction paradigms, various interaction paradigms ranging from time-sharing computers, through the WIMP and web, to ubiquitous and context-aware computing.
5.	Interaction design basics, introduction to design process, user focus, scenarios, Navigation design, screen layout, iteration and prototyping
6.	HCI in software process, Software life Cycle, usability engineering, iterative design and prototyping, Design rational
7.	Design rules, Principles to support usability, standards, guidelines, Golden rules and Heuristics, HCI patterns
8.	Evaluation techniques, Goals of evaluation, evaluation through expert analysis and user participation, selection of evaluation method
9.	Universal design introduction, Universal design principles, multimodal interaction,
10.	User support, requirements for user support, approaches to user support, adaptive help systems, designing user support systems
11.	Cognitive models, introduction, Goal and task hierarchies, Linguistic models, The challenges of display based systems, physical and device models, Cognitive architectures
12.	Socio-organizational issues and stakeholders requirements,
13.	Communication and collaboration models, Face to face, text based communication, Group ware
14.	Task analysis, task decomposition, knowledge based analysis, entity relationship based Techniques, source of information and data collection, uses of task analysis
15.	Dialogue notations and design, diagrammatic, textual and semantic notations
16.	Modeling rich interactions

Course Name: Parallel and Distributed Computing

Credit Hours: 3 (2-3)

Course Introduction:

This course covers a broad range of topics related to parallel and distributed computing, including parallel and distributed architectures and systems, parallel and distributed programming paradigms, parallel algorithms, and scientific and other applications of parallel and distributed computing.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Learn about parallel and distributed computers.	C2 (Understand)	2
CLO-2	Write portable programs for parallel or distributed architectures using Message-Passing Interface (MPI) library	C4 (Apply)	4
CLO-3	Analyze complex problems with shared memory programming with openMP.	C5 (Analyze)	3

Course Outline:

Asynchronous/synchronous computation/communication, concurrency control, fault tolerance, GPU architecture and programming, heterogeneity, interconnection topologies, load balancing, memory

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consistency model, memory hierarchies, Message passing interface (MPI), MIMD/SIMD, multithreaded programming, parallel algorithms and architectures, parallel I/O, performance analysis and tuning, power, programming models (data parallel, task parallel, process-centric, shared/distributed memory), scalability and performance studies, scheduling, storage systems, synchronization, and tools (Cuda, Swift, Globus, Condor, Amazon AWS, OpenStack, Cilk, gdb, threads, MPICH, OpenMP, Hadoop, FUSE).

Reference Materials:

- Distributed Systems: Principles and Paradigms, A. S. Tanenbaum and M. V. Steen, Prentice Hall, latest Edition.
- Distributed and Cloud Computing: Clusters, Grids, Clouds, and the Future Internet, K Hwang, J Dongarra and GC. C. Fox, Elsevier, latest Edition.

	Course Name: Parallel and Distributed Computing
Week #	**Week wise distribution**
1.	Introduction to parallel and distributed computing, Asynchronous/synchronous computation/communication
2.	concurrency control
3.	fault tolerance, GPU architecture and programming
4.	memory hierarchies
5.	Message passing interface (MPI), MIMD/SIMD
6.	multithreaded programming
7.	parallel algorithms and architectures
8.	parallel I/O, performance analysis and tuning
9.	power, programming models (data parallel, task parallel, process-centric, shared/distributed memory)
10.	scalability and performance studies
11.	Scheduling parallel tasks
12.	storage systems
13.	Introduction to cloud computing
14.	synchronization
15.	Introduction to Parallel and distributed computing tools (Cuda, Swift, Globus, Condor)
16.	Introduction to Parallel and distributed computing tools (Amazon AWS, OpenStack, Cilk, gdb, threads, MPICH, OpenMP, Hadoop, FUSE).

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Course Name: Advanced Database Management Systems
Credit Hrs: 3 (2, 3)

Course Introduction:
Advanced Database Management Systems is an extension to the "Database Systems" course. The aim of the course is to enhance the previous knowledge of database systems by deepening the understanding of the theoretical and practical aspects of the database technologies and showing the need for distributed database technology to tackle deficiencies of the centralized database systems. Moreover, it focuses on introducing the basic principles and implementation techniques of distributed database systems and exposes emerging research issues in database systems and application development.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO 1	Understanding advanced data models, technologies and approaches for building distributed database systems.	C2 (Understand)	2
CLO 2	Applying the models and approaches in order to become enabled to select and apply appropriate methods for a particular case	C3 (Apply)	3
CLO 3	To develop a database solution for a given scenario/ challenging problem in the domain of distributed database systems	C3 (Apply) C4 (Design)	3, 4, 6

Course Outline:
Introduction to advanced data models such as object relational, object oriented, File organizations concepts, Transactional processing and Concurrency control techniques, DB Recovery techniques, Query processing and optimization, Database Programming (PL/SQL, T-SQL or similar technology), DB Integrity and security, DB Integrity and security, Database Administration (Role management, managing database access, views), Physical database design and tuning , Distributed database systems, Emerging research trends in database systems, MONGODB, NO SQL, Cloud adoption, AI integration (or similar technologies).

Reference Books:
- Modern Database Management, Jeffrey A. Hoffer et al., Latest Edition.
- Database Systems: A Practical Approach to Design, Implementation and Management. Thomas Connolly and Carolyn Beg, Latest Edition.
- Database Management Systems by Raghu Ramakrishnan, Johannes Gehrke, Latest Edition
- Database System Concepts by Avi Silberschatz, Henry F. Korth and S. Sudarshan Latest Edition.
- Database Systems: The Complete Book, Latest Edition by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom.

Course Name: Advanced Database Management Systems	
Week#	**Week wise distribution**
1.	Introduction to advanced data models such as object relational, object oriented
2.	File organizations concepts
3.	Transactional processing and Concurrency control techniques

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4.	DB Recovery techniques
5.	Query processing and optimization
6.	Query processing and optimization
7.	Database Programming (PL/SQL, T-SQL or similar technology)
8.	Database Programming (PL/SQL, T-SQL or similar technology) (2)
9.	MID TERM EXAMS
10.	DB Integrity and security
11.	Database Administration (Role management, managing database access, views)
12.	Physical database design and tuning
13.	Distributed database systems
14.	Emerging research trends in database systems
15.	MONGODB
16.	NO SQL, Cloud adoption
17.	AI integration (or similar technologies)
18.	FINAL TERM EXAMS

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Course Name: Theory of Automata
Credit Hours: 3(3-0)

Course Introduction:
The course introduces some fundamental concepts in automata theory and formal languages including grammar, finite automaton, regular expressions, formal language, pushdown automaton, and Turing machine. The course introduces the basic models of computation as foundation of many branches of computer science, e.g. compilers, software engineering, concurrent systems, etc.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Explain and manipulate the different concepts in automata theory and formal languages such as formal proofs, automata, regular expressions, Turing machines etc.	C2 (Understand)	2
CLO-2	Prove properties of languages, grammars and automata with rigorously formal mathematical methods	C2 (Understand)	2
CLO-3	Design of automata, RE and CFG	C3 (Apply)	4
CLO-4	Transform between equivalent NFAs, DFAs and REs	C3 (Apply)	4
CLO-5	Define Turing machines performing simple tasks	C2 (Understand)	2
CLO-6	Differentiate and manipulate formal descriptions of languages, automata and grammar with focus on regular and context-free languages, finite automata and regular expressions.	C3 (Apply)	4

Course Outline:
Finite State Models: Language definitions preliminaries, Regular expressions/Regular languages, Finite automata (FAs), Transition graphs (TGs), NFAs, Kleene's theorem, Transducers (automata with output), Pumping lemma and non-regular language Grammars and PDA: CFGs, Derivations, derivation trees and ambiguity, Simplifying CFLs, Normal form grammars and parsing, Decidability, Context sensitive languages, grammars and linear bounded automata (LBA), Chomsky's hierarchy of grammars Turing Machines Theory: Turing machines, Post machine, Variations on TM, TM encoding, Universal Turing Machine, Defining Computers by TMs.

Reference Materials:
- Introduction to computer theory, Daniel I. A. Cohen, Latest Edition
- Automata, Computability and Complexity: Theory and Applications, by Elaine Rich, 2011
- An Introduction to Formal Languages and Automata, by Peter Linz, Latest edition, Jones and Bartlett Publishers, Latest Edition
- Theory of Automata, Formal Languages and Computation, by S. P. Eugene, Kavier, Latest Edition

Course Name: Theory of Automata and Formal Languages	
Week #	**Week wise distribution**

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1.	Finite State Models: Language definitions preliminaries, Regular expressions/Regular languages
2.	Finite automata FAs, Transition graphs TGs)
3.	NFAs, Kleene's theorem
4.	Transducers automata with output
5.	Pumping lemma and non-regular language Grammars
6.	PDA
7.	Context Free Grammar, Derivations, derivation trees and ambiguity
8.	Simplifying CFLs, Normal form grammars and parsing
9.	Decidability, Context sensitive languages
10.	Grammars and linear bounded automata LBA
11.	Chomsky's hierarchy of grammars Turing Machines Theory
12.	Turing machines
13.	Post machine, Variations on TM
14.	TM encoding, Universal Turing Machine
15.	Defining Computers by TMs
16.	Wrap up work and presentations

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Course Name: Compiler Construction

Credit Hours: 3(2-3)

Course Introduction:

This course introduces the basic techniques for Compiler Construction. The course also introduces the theory and tools that can be standardly employed to perform syntax-directed translation of a high-level Programming language into an executable code.

#	CLOs	BT Level	PLOs
CLO1	Understand the basic techniques used in compiler construction	C2(Understand)	2
CLO2	Understand the basic data structures used in compiler construction	C2(Understand)	2
CLO3	Design and implement a compiler.	C5(Create)	4, 6
CLO4	Implementation of tokenization and parsing	C3 (Apply)	3, 4

Course Outlines:

Introduction to interpreter and compiler. Compiler techniques and methodology; Organization of compilers; Lexical and syntax analysis; Parsing techniques. Types of parsers, top-down parsing, bottom up parsing, Type checking, Semantic analyzer, Object code generation and optimization, detection, and recovery from errors. Symbol Table.

Reference books

- Watson, D. A Practical Approach to Compiler Construction (Latest ed.). Springer.
- Compilers: Principles, Techniques, and Tools by Alfred V. Aho, RaviSethi, Jeffrey D. Ullman, Contributor Jeffrey D. Ullman, Addison-Wesley Pub. Co., Latest Edition, Original from the University of Michigan
- Mogensen, T. Æ. Introduction to compiler design (Latest ed.). Springer Science and Business Media.
- Dave, P. H., and Dave, H. B. Compilers: Principles and Practice (Latest ed.). Pearson Education India
- Puntambekar, A. A. Principles of compiler design (Latest ed.). Technical Publications Cooper, K., and Torczon, L Engineering a compiler (Latest ed.). Elsevier.

	Course Contents
Week	**Topics Covered**
1	Introduction to interpreter and compiler.
2	Compiler techniques and methodology.
3	Organization of compilers.
4	Lexical Analysis
5	Regular Expression

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6	Finite Automata
7	Syntax Analysis
8	top-down parsing,
9	bottom-up parsing,
10	Type checking
11	Semantic analyzer,
12	Syntax Directed Translation
13	Object code generation
14	Code optimization
15	Detection and recovery from errors
16	Wrap up work and presentations

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Course Name: Computer Architecture
Credit Hours: 3 (2-3)

Course Description:
This course is about a thorough understanding of the basic structure and operation of a digital Computer. To discuss in detail the operation, The course covers the basic principles of micro-computer architecture, its organization, operation, and performance. It also deals with embedded systems and memory management The course includes the cache memory, its organization and mapping functions, this course will help and prepare the students to take Advance courses in Computer Architecture in future.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand the basic model of a computer including the Datapath, control, memory, and I/O components	C2(Comprehension)	2
CLO-2	Understand the memory hierarchy including caches and virtual memory	C2(Comprehension)	2
CLO-3	Understand computer performance measures and their estimation	C2(Comprehension)	2
CLO-4	Understand pipelined architectures including data-paths and hazards	C2(Comprehension)	2
CLO-5	write a fully functional, stand-alone medium size assembly language program	C3(Apply)	4, 5

Course Outlines:
Fundamentals of Computer Design including performance measurements and quantitative principles, principles of Instruction Set Design, Operands, addressing modes and encoding, pipelining of Processors: CPU Simulators, Issues and Hurdles, exception handling features, Instruction-Level Parallelism and Dynamic handling of Exceptions, Memory Hierarchy Design, Cache Design, Performance Issues and improvements, Main Memory Performance Issues, Storage Systems, Multiprocessors and Thread Level Parallelism. Case Studies.

Reference Materials:
- Computer Architecture: A Quantitative Approach by Hennessy and Patterson, Morgan and Kauffman Series Latest Edition .
- Computer Organization and Design: The Hardware/Software Interface by Patterson and Hennessy, Morgan and Kauffman Series latest Edition .
- Computer Organization and Architecture Designing for performance by William Stallings, Latest Edition, Prentice Hall Inc.
- The 8088 and 8086 Microprocessors by Walter A. Triebel, Avtar Singh fourth edition Prentice Hall

Course Name: Python Programming	
Week #	**Week wise distribution**
1.	The design of computer systems and components.
2.	8086 Microprocessor
3.	Instruction set design (ISA)
4.	Addressing modes and ISA dimensions
5.	Control structures and microprogramming;
6.	RISC and CISC architectures

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7.	Quantitative principles of Computer Design
8.	Interconnection Structures
9.	Memory hierarchies and Management
10.	Cache, levels, placement, and optimization
11.	Virtual memory and virtual machines,
12.	Interrupts and I/O structures.
13.	Pipelining of processor Issues and Hurdles, CPU Simulators
14.	Exception handling, Parallelism (ILP, DLP, TLP)
15.	Multiprocessor Systems.
16.	Wrap up work and presentations

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Domain Elective Subjects

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Course Name: Python Programming
Credit Hours: 3 (2-3)

Course Description:
The Python Programming course is designed for beginners and those looking to enhance their programming skills using Python. It covers fundamental concepts such as variables, data types, operators, control structures, and error handling techniques. Students will learn to manipulate strings and lists, utilize built-in functions, and explore graphics programming with the Turtle library. The curriculum also includes data manipulation using the Pandas library and practical sessions on file handling with JSON and CSV formats. The course culminates in a capstone project, allowing students to apply their knowledge to build a functional Python application, equipping them with the skills needed to tackle real-world programming challenges.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Demonstrate a solid understanding of basic programming concepts by writing and executing Python scripts.	C1 (Knowledge)	2
CLO-2	Apply Python libraries, such as Pandas, to manipulate, analyze, and visualize data, and handle different file formats like JSON and CSV.	C2(Understand)	3
CLO-3	Design and implement a functional Python application through a capstone project, showcasing the ability to solve real-world problems using programming techniques and best practices.	C3(Apply)	3,4

Course Outlines:
Introduction to Python, Downloading and Installation of Python IDE (PyCharm), Anatomy of Python Program, Guidelines for Creating Script, Programming Errors, Variables and Operators, Variables, Variable Names, Creating Variables, Variable Casting, Python Assignment Operators, Python Operators, Type Conversion vs Type Casting, Strings in Python, Combine vs Repeat Strings, Strings Built-in Methods, Python Lists, List Items, List Ordered, Python Collections (Arrays), Forward vs Backward Accessing of List, Changing List Item, Adding Element into List, Removing List Item, List Slicing, Membership Operator in Python, Python Conditional Statement, If Statement, Else Statement, Elsif Statement, Use of Logical Operator with If Condition, Python Loops, The For Loop, The While Loop, Python Functions, Variable Scope, Parameters or Arguments, Number of Arguments, Arbitrary Arguments (args), Python Try Except, Working with Graphics, Introduction to Turtle, Plotting using Turtle, Python Pandas Library.

Reference Materials:
- Lutz, Mark. Learning Python, O'Reilly Media, Latest Edition
- Ramalho, Luciano. Fluent Python: Clear, Concise Programming. O'Reilly Media, Latest Edition
- Sweigart, Al. Automate the Boring Stuff with Python: Practical Programming for Total Beginners.
- Matthes, Eric. Python Crash Course: A Hands-On, Project-Based Introduction to Programming.

Course Name: Python Programming	
Week #	**Week wise distribution**
1.	Introduction to Python Programming Concepts, Overview of Python Programming Languages and Their Evolution, Significance of Python in Contemporary Software Development.,

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2.	Downloading and Installation of Python IDE (PyCharm), Lab: Writing and running a simple Python programs.
3.	Anatomy of Python Program, Guidelines for Creating Scripts, Programming Errors, Debugging Techniques, Lab: Writing simple scripts and exploring programming errors.
4.	Variables and Operators, Variables, Variable Names, Creating Variables, Variable Casting, Python Assignment Operators, Lab: Practice with variables and assignment operators.
5.	Python Operators, Type Conversion vs Type Casting, Operator Precedence and Associativity, Lab: Programs with operators and type conversion.
6.	Strings in Python, Combine vs Repeat Strings, Strings Built-in Methods, Lab: String operations and built-in string methods.
7.	Python Lists, List Items, List Ordered, Python Collections (Arrays), Lab: Creating and manipulating lists in Python.
8.	Forward vs Backward Accessing of List, Changing List Items, Adding Elements into List, Removing List Items, List Slicing, Lab: List manipulation and slicing.
9.	Membership Operator in Python, Python Conditional Statement (If, Else, Elif), Lab: Writing programs with conditional statements.
10.	Use of Logical Operators with If Condition, Python Loops (For, While), Lab: Practice with loops and logical operators.
11.	Python Functions, Variable Scope, Parameters or Arguments, Number of Arguments, Arbitrary Arguments (*args), Lab: Creating functions and exploring argument types.
12.	Python Try Except, Writing Robust Code with Error Handling, Function Recursion, Lab: Working with try-except blocks and recursion.
13.	Working with Graphics, Introduction to Turtle, Plotting using Turtle, Introduction to Object-Oriented Programming Concepts (Basics), Lab: Plotting shapes using Turtle.
14.	Python Pandas Library, Data Manipulation with Pandas, Lab: Data frames and basic data analysis with Pandas.
15.	File Handling in Python (Reading/Writing Files), Working with JSON and CSV files, Lab: File input/output scripts.
16.	Exercises (Practicals, Fill in the blanks, Multiple Choice Questions), Capstone Project: Building a Small Python Application, Lab: Final project presentations and assessments.

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Course Name: Digital Image Processing
Credit Hours: 3 (2-3)

Course Introduction:
This course provides an in-depth understanding of digital image processing techniques, focusing on image enhancement, segmentation, registration, and transformation methods.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand the foundational concepts of digital image processing, including image representation, sampling, quantization, and pixel relationships	C2 (Understand)	1
CLO-2	Identify appropriate image enhancement and segmentation techniques	C3 (Identify)	2
CLO-3	Apply image enhancement, segmentation, and registration techniques to real-world image processing problems.	C3 (Apply)	5
CLO-4	Examine and evaluate the effectiveness of morphological operations, image transforms, and registration methods in image analysis tasks.	C4 (Examine)	3

Course Outline:
Introduction: Elements of digital image processing, Image model, Sampling and quantization, Relationships between pixels, Image Enhancement: Enhancement by point processing, Spatial filtering, Enhancement in the frequency domain, Colour Image Processing, image Segmentation: Discontinuity detection, Edge linking and boundary detection, Treshholding, Region oriented segmentation, Use of motion for segmentation, Image Registration: Introduction to image registration, Techniques of image registration, Representation and Description: Boundary description, Regional description, Morphological Image Processing: Dilation and Erosion, Opening and Closing, Some basic morphological algorithms, Extensions to gray level images, Image Transforms: Discrete Fourier Transform, Discrete Cosine Transform, Haar Transform, Hadamard Transform

Reference Material:
- Digital Image Processing, R. C. Gonzalez and R. E. Woods, Latest Edition, Prentice Hall
- Understanding Digital Signal Processing by Richard G. Lyons, Prentice Hall,Latest Edition

Course Name: Object Oriented Programming	
Week #	**Week wise distribution**
1	Introduction to Digital Image Processing
2	Sampling and Quantization
3	Image Enhancement by Point Processing
4	Spatial Filtering
5	Enhancement in the Frequency Domain
6	Color Image Processing
7	Image Segmentation - Discontinuity Detection
8	Image Segmentation - Edge Linking and Boundary Detection
9	Thresholding Techniques
10	Region-Oriented Segmentation
11	Use of Motion for Segmentation

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12	Introduction to Image Registration
13	Image Registration Techniques
14	Morphological Image Processing - Dilation and Erosion
15	Opening and Closing, Morphological Algorithms
16	Image Transforms

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Course Name: Visual Programming in Java
Credit Hours: 3 (2-3)

Course Introduction:
The course aims to focus on Visual Programming in Java that spans basic to advanced Java programming concepts, focusing on both core Java and visual programming using frameworks like JavaFX and Swing.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLO
CLO-1	Understand basic Java.	C2 (Understand)	2
CLO-2	Proficiency in the effective use of the Java Collections Framework	C3 (Demonstrate)	2, 3
CLO-3	Java applications integrated with databases using JDBC, performing data management tasks such as CRUD operations	C3 (Apply)	5
CLO-4	Create comprehensive, real-world desktop applications that incorporate advanced programming techniques in software development	C4 (Construct)	4

Course Outline:
Overview of Java, Installation of JDK and IDE (Eclipse/NetBeans/IntelliJ), First Java Program: Hello World, Data Types, Variables, and Operators, Control Flow Statements: if, else, switch, loops (for, while, do-while), Introduction to Methods and Functions, Introduction to Classes and Objects, Constructors and Initialization, Inheritance and Polymorphism, Abstract Classes and Interfaces, Packages and Access Modifiers, Exception Handling, Overview of Java Collections: Lists, Sets, Maps ArrayList, LinkedList, HashSet, TreeSet, HashMap, TreeMap, Overview of Java GUI Frameworks (AWT, Swing, JavaFX), Introduction to Java Swing, Creating Simple WindowsLayout Managers, Event Handling in Swing, Common Swing Components: Buttons, Labels, TextFields, TextAreas, Introduction to JavaFX, Setting up a JavaFX Project, JavaFX Application Structure, Scene Graph and Nodes, Iterators, For-Each Loop, and Streams API (basic introduction), JavaFX UI Components: Buttons, TextFields, Labels, Menus, Handling Events in JavaFX, Layout Management in JavaFX: VBox, HBox, BorderPane, GridPane, FlowPane, CSS Styling in JavaFX, Handling Multiple Scenes and Stages, Using FXML to Build UIs, Introduction to JDBC (Java Database Connectivity), Setting up MySQL/SQLite Database, Connecting Java Applications to a Database, Performing CRUD Operations (Create, Read, Update, Delete), Prepared Statements and ResultSets, Multithreading and Concurrency, Creating and Running Threads, Synchronization and Locks.

Reference Materials: (or use any other standard and latest books)
- Java: How to Program, Latest Edition by Paul Deitel
- Core Java Volume I – Fundamentals" by Cay S. Horstmann and Gary Cornell
- JavaFX 8: Introduction by Example" by Carl Dea
- ava Database Programming with JDBC and JDBC-ODBC" by Alan Bateman.
- Object Oriented Programming in C++, 3rd Edition by Robert Lafore.

Course Name: Visual Programming in Java	
Week #	**Week wise distribution**
1	Overview of Java, Installation of JDK and IDE (Eclipse/NetBeans/IntelliJ), First Java Program: Hello World

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2	Data Types, Variables, and Operators, Control Flow Statements: if, else, switch, loops (for, while, do-while), Introduction to Methods and Functions
3	Abstract Classes and Interfaces, Packages and Access Modifiers, Exception Handling
4	Overview of Java Collections: Lists, Sets, Maps ArrayList, LinkedList, HashSet, TreeSet, HashMap, TreeMap
5	Overview of Java GUI Frameworks (AWT, Swing, JavaFX), Introduction to Java Swing
6	Creating Simple WindowsLayout Managers, Event Handling in Swing, Common Swing Components: Buttons, Labels, TextFields, TextAreas
7	Introduction to JavaFX, Setting up a JavaFX Project,
8	JavaFX Application Structure, Scene Graph and Nodes, Iterators,
9	For-Each Loop, and Streams API (basic introduction), JavaFX UI Components: Buttons, TextFields, Labels, Menus,
10	Handling Events in JavaFX, Layout Management in JavaFX: VBox, HBox, BorderPane, GridPane, FlowPane
11	CSS Styling in JavaFX, Handling Multiple Scenes and Stages, Using FXML to Build UIs,
12	Introduction to JDBC (Java Database Connectivity), Setting up, MySQL/SQLite Database
13	Setting up MySQL/SQLite Database, Connecting Java Applications to a Database, Performing CRUD Operations (Create, Read, Update, Delete), Prepared Statements and Result Sets,
14	polymorphism, Multithreading and Concurrency, Creating and Running Threads
15	exception handling.
16	Wrap up work and presentations

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Course Name: Web Desig and Development
Credit Hours: 3 (2-3)

Course Introduction:
Web Technologies courses cover the fundamentals of building and maintaining websites, including HTML, CSS, JavaScript, and more. Gain the skills needed to create engaging web experiences and pursue a career in web development.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	To know the fundamentals of web application architecture and web programming.	C2 (Understand)	**2**
CLO-2	To apply a structured approach to identifying needs, interests, and functionality of a website.	C2 (Understand)	**3**
CLO-3	To design and implement an interactive web site(s) about issues of usability, accessibility, and internationalization.	C4 (Apply)	**4**

Course Outline:
History and Advantages and Disadvantages of Internet, Web Server, Web Browser, Web Clients, and Search Engines, Client-Server Architecture, Types and Categories of Websites, Creation and Basic Structure of HTML Document, HTML Tags (Headings , Paragraphs, Line Break, Horizontal Line, Font, Preformatted Text, Lists, Images, Tables, Hyperlink, Fame, and Form), CSS, Inserting JavaScript Code in HTML Document, JavaScript Constructs (Variables and Rules of Naming Variables, Operators, Type Casting, Decision Control Structures, Loops, Function, Array, and DOM), Installing and Configuring Apache and PHP, Creating PHP File, Overview of Variables and Constants, Output Statement in PHP, Passing Variables Between Pages (URL, Sessions, Cookies, and Forms), Accessing and Using Database in PHP, and Database, AJAX, Introduction to Service Oriented Architecture and Web Services, Designing and Implementing Web Services with SOAP and JSON.

Reference Materials:

- Duckett, J.. Web Design with HTML, CSS, JavaScript and jQuery Set (Latest ed.). Wiley Publishing
- Nixon, R. PHP: 20 Lessons to Successful Web Development (Latest ed.). McGrawHill Education Group
- Nixon, R. Learning PHP, MySQL, JavaScript, and CSS: A step-by-step guide to creating dynamic websites (Latest ed.). O'Reilly Media, Inc

Course Name: Web Desig and Development	
Week #	**Week wise distribution**
1.	History and Advantages and Disadvantages of Internet, Web Server, Web Browser, Web Clients, and Search Engines
2.	Client-Server Architecture, Types and Categories of Websites, Creation and Basic Structure of HTML Document,

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3.	HTML Tags (Headings, Paragraphs, Line Break, Horizontal Line, Font, Preformatted Text, Lists, Images, Tables, Hyperlink, Fame, and Form)
4.	CSS,
5.	Inserting JavaScript Code in HTML Document,
6.	JavaScript Constructs (Variables and Rules of Naming Variables)
7.	JavaScript Constructs (Operators, Type Casting)
8.	JavaScript Constructs (Decision Control Structures, Loops, Function, Array, and DOM)
9.	Installing and Configuring Apache and PHP,
10.	Creating PHP File,
11.	Overview of Variables and Constants, Output Statement in PHP
12.	Passing Variables Between Pages (URL, Sessions)
13.	Passing Variables Between Pages (Cookies, and Forms)
14.	Accessing and Using Database in PHP, and Database, AJAX,
15.	Introduction to Service Oriented Architecture and Web Services,
16.	Designing and Implementing Web Services with SOAP and JSON

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Course Name: Web Engineering

Credit Hours: 3 (2-3)

Course Introduction:

This course introduces a structured methodology utilized in software engineering to web development projects. This course examines systematic, disciplined and quantifiable approaches to develop high-quality, reliable and usable web applications. The course addresses the concepts, methods, technologies, and techniques of developing web sites that collect, organize and expose information resources.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Discuss how web standards impact software development.	C2 (Understand)	2
CLO-2	Describe the constraints that the web puts on developers.	C2 (Understand)	2
CLO-3	Design and implement a simple web application	C3 (Design)	4
CLO-4	Analyze an existing web application against a current web standard	C5 (Analyze)	3

Course Outline:

Web programming languages (e.g., HTML5, CSS 3, Java Script, PHP/JSP/ASP.Net), Design principles of Web based applications, Web platform constraints, Software as a Service (SaaS), Web standards, Responsive Web Design, Web Applications, Browser/Server Communication, Storage Tier, Cookies and Sessions, Input Validation, Full stack state management, Web App Security - Browser Isolation, Network Attacks, Session Attacks, Large scale applications, Performance of Web Applications, Data Centers, Web Testing and Web Maintenance.

Reference Materials

- 1. Web Engineering, Rajiv Chopra, Prentice-Hall of India, Latest Edition.
- Web Engineering, Emilia Mendes and Nile Mosley, Springer Verlag, Latest Edition..
- Web Engineering: A Practitioners' Approach, Roger S. Pressman, McGraw Hill, Latest Edition..
- Dynamic HTML: The Definitive Reference: A Comprehensive Resource for XHTML, CSS, DOM, JavaScript Late.st Edition.
- JavaScript: The Definitive Guide, Latest Edition, David Flanagan. O'Reilly Media.

Course Name: Web Engineering	
Week #	**Week wise distribution**
1.	Introduction to web technologies, web engineering
2.	Web programming languages (e.g., HTML5, java Script, CSS,PHP/JSP/ASP.Net)
3.	Design principles of Web based applications
4.	Web platform constraints
5.	Software as a Service (SaaS), Platform as a Service (Paas)

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6.	Web standards
7.	Responsive Web Design
8.	Web Applications
9.	Browser/Server Communication
10.	Storage Tier, Cookies and Sessions
11.	Input Validation, Full stack state management
12.	Web App Security - Browser Isolation
13.	Network Attacks, Session Attacks
14.	Large scale applications
15.	Performance of Web Applications, Data Centers
16.	Web Testing and Web Maintenance.

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Course Name: Numerical Analysis
Credit Hours: 3 (2-3)

Course Introduction:
This course studies algorithms that use numerical approximation for the problems of mathematical analysis. Numerical methods, based upon sound computational mathematics, are the basic algorithms underpinning computer predictions in modern systems science.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand the basics of numerical analysis, algorithms that use numerical approximation and mathematical computations.	C2 (Understand)	2
CLO-2	Understand numerical methods based upon sound computational mathematics	C2 (Understand)	2
CLO-3	Apply basic algorithms underpinning computer predictions in modern systems science.	C4 (Apply)	4

Course Outline:
The concepts of efficiency, reliability and accuracy of a method; Minimizing computational errors; Theory of Differences, Difference Operators, Difference Tables, Forward Differences, Backward Differences and Central Differences. Mathematical Preliminaries, Solution of Equations in one variable, Interpolation and Polynomial Approximation, Numerical Differentiation and Numerical Integration, Initial Value Problems for Ordinary Differential Equations, Direct Methods for Solving Linear Systems, Iterative Techniques in Matrix Algebra, Solution of non-linear equations.

Reference Materials:
- Numerical Methods in Scientific Computing by Germund, D. Åke, B.
- Numerical Methods for Scientific Computing by J. H. Heinbockel.
- Numerical Analysis by I. A. Khubaza.
- Numerical Analysis and Programming by Shan S Kuo.
- Numerical Analysis by Berden, F.
- Numerical Analysis by Gerald.

Course Name: Numerical Analysis	
Week#	**Week wise distribution**
1.	The concepts of efficiency, reliability and accuracy of a method; Minimizing computational errors; Theory of Differences, Difference Operators, Difference Tables
2.	Forward Differences, Backward Differences and Central Differences
3.	Mathematical Preliminaries
4.	Mathematical Preliminaries
5.	variable
6.	Solution of Equations in one

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7.	Interpolation and
8.	Polynomial Approximation
9.	Numerical
10.	Differentiation and Numerical Integration
11.	Initial Value Problems for Ordinary Differential Equations
12.	Direct Methods for Solving
13.	Linear Systems
14.	Iterative Techniques in Matrix Algebra
15.	Solution of non-linear equations
16.	Wrap up work and presentations

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Course Name: Cloud Computing
Credit Hours: 3 (2-3)

Course Introduction:
Cloud computing is a model for delivering computing services over the internet, providing on-demand access to a shared pool of resources (e.g., servers, storage, applications). Cloud computing enables flexible, on-demand access to computing resources, transforming the way businesses operate and innovate.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Explain the core issues of cloud computing such as security, privacy, consistency and interoperability	C2 (Understand)	2
CLO-2	Develop and deploy cloud applications using popular cloud platforms.	C2 (Understand)	2
CLO-3	Compare the key trade-offs between multiple design approaches used for cloud systems.	C5 (Apply)	3, 4

Course Outline:
Datacenter Architectures, Cloud Stack , Technology Trends, Consistency, Availability, Partitions, Cluster File Systems, Data-flow Computation Frameworks, Key-Value Store and Interactive Query Systems, Big Data in the Clouds, Geographic distributed Storage, Programming Languages for the Cloud, DBases in the Cloud, InMemory Frameworks, Google file system, Hadoop file system, MapReduce, Clouds Networking: topologies, Networking: Traffic Management, Networking: Transport Protocol Improvements, Security, Scheduling and Resource Management in clouds, Software Level Agreements.

Reference Materials:
- Cloud Computing Implementation, Management, and Security by John W.Rittinghouse and James F. Ransome, Taylor and Francis Grou, Latest Edition
- Cloud Computing Explained: Implementation Handbook for Enterprises by John Rhoton, Latest Edition
- Cloud Computing and SOA Convergence in Your Enterprise: A Step-by- Step Guide by David S, Latest Edition
- Cloud Computing Bible by Barrie Sosinsky, Wiley; Latest Edition.

Course Name: Cloud Computing	
Week #	**Week wise distribution**
1.	Introduction to Cloud Computing, Objective
2.	Data center architectures, cloud stack components
3.	(IaaS, PaaS, SaaS), and emerging technology trends.
4.	CAP Theorem, explore cluster file systems (HDFS, GFS)
5.	Data-flow computation frameworks (MapReduce, Spark)
6.	Deep dive into cloud storage solutions, explore key-value stores
7.	Interactive query systems, and big data considerations.
8.	Explore various database models (relational, NoSQL) in cloud environments

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9.	Memory frameworks, Managed database services.
10.	Cloud networking topologies, explore traffic management techniques
11.	Transport protocol improvements in the cloud.
12.	Security threats and challenges in cloud environments
13.	Security best practices, compliance requirements.
14.	Cloud resource management concepts, scheduling algorithms
15.	Service Level Agreements (SLAs)
16.	Programming languages and frameworks for developing cloud-native applications.

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Course Name: Cyber Security
Credit Hours: 3(2-3)

Course Introduction:
This course provides students an introduction to common cyber security threats, vulnerabilities, and risks related to web applications, networks, software and mobile applications. The course provides basic concepts and terminology used in the information and cyber security fields. Moreover, it will also enable students to differentiate between the various forms of malware and how they affect computers and networks.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	To be able to identify computer system threats	C2 (Understand)	2
CLO-2	To be able to identify Malware attacks, and understand the stages of attack and payloads	C2 (Understand)	2
CLO-3	Implement various cryptographic techniques and simulate attack scenarios	C4 (Apply)	3,4,5

Course Outline:
Introduction to Cyber security; Networks and the Internet; cyber threat landscape; understanding security; information security Principles (Confidentiality, Integrity, Availability); Information Security Terminology; Who are the attackers; Advanced Persistent Threat (APT); Malware, types of malware; Attacks using malware; Malware Attack Lifecycle: Stages of Attack; Social engineering attacks; types of payload; Industrial Espionage in Cyberspace; Basic cryptography; Web application attacks; Database security; Cyber kill chain; Privacy and anonymity; Network security; Software security; Mobile device security; Mobile app security; Cyber Terrorism and Information Warfare; Introduction to Digital Forensics; Digital Forensics Categories.

Reference Materials:
- Computer Security Fundamentals by Chuck Easttom, Latest edition.
- Security+ Guide to Network Security Fundamentals, by Mark Ciampa, Latest Edition
- Security in Computing by C.P. Pfleeger, Prentice-Hall, Latest Edition.

Course Name: Cyber Security	
Week #	**Week wise distribution**
1.	Introduction to Cyber security;
2.	Networks and the Internet;
3.	cyber threat landscape;
4.	understanding security; information security Principles (Confidentiality, Integrity, Availability); Information Security Terminology
5.	Who are the attackers; Advanced Persistent Threat (APT); Malware, types of malware; Attacks using malware; Malware Attack Lifecycle
6.	Stages of Attack; Social engineering attacks; types of payload
7.	Industrial Espionage in Cyberspace
8.	Basic cryptography

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9.	Web application attacks
10.	Database security
11.	Cyber kill chain
12.	Privacy and anonymity; Network security
13.	Software security; Mobile device security; Mobile app security
14.	Cyber Terrorism and Information Warfare
15.	Introduction to Digital Forensics
16.	Digital Forensics Categories

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Course Name: Software Testing and Quality Assurance

Credit Hours: **3 (2-3)**

Course Introduction:

This course explores the goals of quality assurance and quality control activities performed during the life cycle of a software product. It focuses on integrating test processes with agile software development methodologies. Practical exercises give experience of design, specification, execution of tests plus test automation using tools through a mixture of instructor-directed exercises and student research leading to knowledge sharing.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	A focus on the software testing processes at the unit, module, subsystem, and systems levels.	C2 (Understand)	2
CLO-2	Identify parameters for quality software development.	C2 (Understand)	2,3
CLO-3	Knowledge of tools for software testing which may provide essential skills for ensuring that products and services meet industry standards and customer expectations.	C3 (Apply)	4

Course Outline:

Need of software testing and quality assurance, the uniqueness of software quality assurance (SQA), environments for which SQA methods are developed, software errors, faults and failures, classification of the causes of software errors, the need for comprehensive software quality requirements, classifications of software requirements into software quality factors (classic model of software quality factors, suggested by McCall and subsequent models suggested by Deutsch and Willis and by Evans and Marciniak), the components of the software quality assurance i.e pre-project components (contract review etc) and software project life cycle components (reviews, expert onions and software testing etc), formal design reviews (DRs) and peer reviews, software testing, definition and objectives, software testing strategies, software test classifications, white box and black box testing, software testing implementation i.e. the testing process (determining the test methodology phase, planning the tests, test design and test implementation), test case design, automated testing and, alpha and beta site testing.

Reference Materials:

- Software Quality Assurance, From theory to implementation by Daniel Galin latest edition
- Paul Jorgensen, Software Testing, A Craftsman's Approach, Latest Edition.
- Software Security: Building Security In, Latest Edition by Gary McGraw
- Software Engineering, A Practitioner's Approach, Pressman R. S.and Maxim B. R., Latest Edition.

Software Testing and Quality Assurance	
Week #	**Week wise distribution**
1.	Need of software testing and quality assurance
2.	The uniqueness of software quality assurance (SQA), environments for which SQA methods are developed
3.	software errors, faults and failures, classification of the causes of software errors

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4 &5.	The need for comprehensive software quality requirements, classifications of software requirements into software quality factors (classic model of software quality factors, suggested by McCall and subsequent models suggested by Deutsch and Willis and by Evans and Marciniak)
6.	the components of the software quality assurance i.e pre-project components (contract review etc)
7.	software project life cycle components (reviews, expert onions and software testing etc)
8.	formal design reviews (DRs) and peer reviews
9.	software testing, definition and objectives, software testing strategies
10.	software test classifications, white box and black box testing
11&12	software testing implementation i.e. the testing process (determining the test methodology phase, planning the tests, test design and test implementation
13.	test case design
14.	automated testing
15.	alpha and beta site testing
16.	Wrap up work and presentations

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Course Title: Mobile App Development
Credit Hours: 3 (2-3)

Course Description:
This course covers the fundamentals of Android application development, including setting up the Android SDK, creating and managing Android apps, and understanding app components such as activities, fragments, and intents. Students will learn how to build user interfaces, handle notifications, persist and share data, and integrate system features like SMS, emails, and maps. Additional topics include game development, web services, threading, and publishing apps. The course also addresses advanced concepts such as Android architecture, material design, Firebase, multimedia handling, security, and dependency injection.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Demonstrate proficiency in setting up the Android development environment and developing basic Android applications using core components like activities, intents, and fragments.	C3(Apply)	4
CLO-2	Design and implement responsive user interfaces for Android applications that adapt to different screen orientations and user interactions.	C6(Create)	4, 5
CLO-3	Apply data persistence and sharing techniques in Android using local databases (SQLite, Room) and web services (HTTP, Firebase).	C3(Apply)	3
CLO-4	Develop and publish Android applications incorporating system functionalities such as SMS messaging, multimedia handling, and secure data storage, while ensuring performance and security.	C6(Create)	4

Course Outlines:
installing and configuring the Android SDK Manager, creating Android applications, anatomy of an Android application, Eclipse, fragments, calling built-in applications using intents, displaying notifications, components of a screen, adapting to display orientation, managing changes to screen orientation, utilizing the Action Bar, creating the user interface, listening for UI notifications, views, user preferences, persisting data, sharing data, sending SMS messages, getting feedback, sending emails, displaying maps, consuming web services using HTTP, accessing web services, creating services, threading, Android games development, publishing Android applications, handling telephone calls, and fonts. Additional relevant topics include Android Studio, Android architecture (activities, services, broadcast receivers, content providers), RecyclerView and ListView, material design principles, database management (SQLite, Room), content providers, sensors and location services, Bluetooth and NFC communication, handling permissions, Google Play Services integration, Firebase integration, testing Android applications (JUnit, Espresso), handling multimedia (audio, video, camera), notifications (push notifications, local notifications), security in Android applications (encryption, secure storage), and dependency injection (Dagger, Hilt).

Reference Materials:
- Android Programming with Kotlin for Beginners by Horton, John Latest Edition
- Android App Development For Dummies by Burton, Michael. Latest Edition.
- "Android 9 Development Cookbook by Boyer, Rick.
- Android Developers Official Training Courses. Available at: developer.android.com

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Course Name: Mobile App Development	
Week #	**Week wise distribution**
1	Introduction to Android, setting up the Android development environment, installing and configuring Android SDK Manager.
2	Anatomy of an Android application, basic app creation, using Android Studio and Eclipse IDE.
3	Understanding activities and fragments, lifecycle of Android apps, calling built-in applications using intents.
4	User interface design, creating views and layouts, adapting to display orientation, utilizing the Action Bar.
5	Handling UI notifications, managing changes to screen orientation, components of a screen, user preferences.
6	Displaying notifications, sending SMS messages, making phone calls, getting feedback from users.
7	Persisting data, using local storage (SQLite, Room), sharing data between apps.
8	Accessing web services, consuming web services using HTTP, introduction to Firebase integration.
9	Handling multimedia (audio, video, camera), working with maps, displaying location-based data.
10	Android architecture (activities, services, broadcast receivers, content providers), managing permissions.
11	Game development in Android, threading, and asynchronous processing.
12	Inter Introduction to material design principles, working with RecyclerView and ListView.
13	Advanced UI design using Jetpack Compose, working with sensors and location services.
14	Security in Android applications (encryption, secure storage), handling Bluetooth and NFC communication.
15	Testing Android applications (JUnit, Espresso), handling push notifications, Firebase cloud messaging.
16	Publishing Android applications, handling updates, app maintenance, course recap.

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Course Title: **Computer Graphics**
Credit Hours: 3 (2-3)

Introduction: The course introduces the key concepts, techniques and algorithms related to computer graphics.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Ability to understand the goal and applications of computer graphics	C2(Comprehension)	2
CLO-2	Understand coordinate systems and their conversions along with basic algorithms for 2-D drawing primitives	C2(Comprehension)	2
CLO-3	Design and implement algorithms in a computer language	C3 (Apply)	4, 5

Course Outlines:
Graphics hardware. Fundamental algorithms. Applications of graphics. Interactive graphics programming graph plotting, windows and clipping, and segmentation. Programming raster display systems, Differential Line Algorithm, panning and zooming. Raster algorithms and software Scan Converting lines, characters and circles. Scaling, Rotation, Translation, Region filling and clipping. Two and three dimensional imaging geometry (Perspective projection and Orthogonal projection) and transformations. Curve and surface design, rendering, shading, color and animation.

Reference Materials:

- Computer Graphics, Principles and Practice, J. D. Foley, A. van Dam, S. K. Feiner and J. F. Hughes, Addison-Wesley ISBN: 0-201-12110-7.
- Computer Graphics, F.S.Hill, Maxwell MacMillan ISBN: 0-02-354860-6.
- Interactive Computer Graphics: Functional, Procedural and Device-level methods; Peter Burger and Duncan. F. Gillies; Addison-Wesley, (2003)

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Course Name: Computer Graphics	
Week #	**Week wise distribution**
1	Introduction to Computer Graphics and its applications
2	Point, Line Drawing Techniques/algorithms, Circle Drawing Techniques/ algorithms
3	Ellipse and Other Curves, Filled-Area Primitives
4	Mathematics Fundamentals, 2D Transformations
5	Clipping and Drawing example
6	3D Concepts and 3D Transformations
7	Projections, perspective projection and orthogonal projection
8	Triangles and Planes, Triangle Rasterization
9	Lighting
10	Mathematics of Lighting and Shading
11	Mathematics of Lighting and Shading (Continue…)
12	Introduction to OpenGL programing
13	Curves, Space Curves, The Tangent Vector, Bezier Curves
14	Building Polygonal Models of Surfaces, Fractals, Viewing
15	Real-World and OpenGL Lighting
16	Evaluators, curves and Surfaces, Animations.

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Course Name: Object Oriented Analysis and Design
Credit Hours: 3(2-3)

Course Introduction:
This course examines the reasons for the inherent complexity of software construction and presents structured methods to deal effectively with it. The course will focus on the object oriented approach for analysis and design. Students will gain an appreciation of the difference between writing programs and doing analysis and design. Problem formulation and decomposition (analysis) and solution building (design) will be covered. Students will work in small groups, each group having the responsibility for analysis, design and implementation of a software system. Case tools will be used in several stages of the development process.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	To develop a working understanding of formal object-oriented analysis and design processes.	C2 (Understand)	2
CLO-2	To develop an appreciation for and understanding of the risks inherent to large-scale software development.	C2 (Understand)	2
CLO-3	To develop the skills to determine which processes and OOAD techniques should be applied to a given project.	4 (Apply)	4

Course Outline:
Introduction to OOAD and required tolls i.e. Rational Unified Process (RUP), Unified Modeling Language (UML), Business Modeling i.e. Business Objectives, Object Model, (Project Vision Statement), Requirements Engineering, State Diagrams, Activity Diagrams (Project , SRS with UI), Analysis and Design: Fundamentals of Software Architecture, Quality Attributes, Architectural Design and Style, Middleware architecture and technologies, S/W Architectural Process, Documenting Architecture, UML Diagrams, Component Modeling and Package Diagram, Sequence, Collaboration Diagrams and Class Modeling +(Project Architecture, Introduction to J2EE Framework: Components and CORBA Architecture, Software design patterns: Project Demonstrations with Initial Operational Capabilities, Introduction to Software Design Patterns, Principle of Modification of Analysis and Design

Reference Materials:
- Applying UML and Patterns: An Introduction to Object-Oriented Analysis And Design and Iterative Development, Latest Edition, By Craig Larman
- Essential Software Architecture, Springer-Verlag Berlin Heidelberg, By Ian Gorton-Soft Latest Edition.

Course Name: Object Oriented Analysis and Design	
Week #	**Week wise distribution**
1.	Introduction to OOAD and required tolls i.e.
2.	Rational Unified Process (RUP),
3.	Unified Modeling Language (UML)
4.	Business Modeling i.e. Business Objectives, Object Model, +(Project Vision Statement)

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5.	Requirements Engineering, State Diagrams + (Project , SRS with UI)
6.	Requirements Engineering, Activity Diagrams + (Project , SRS with UI)
7.	Analysis and Design: Fundamentals of Software Architecture, Quality Attributes,
8.	Architectural Design and Style
9.	Middleware architecture and technologies,
10.	S/W Architectural Process, Documenting Architecture
11.	UML Diagrams, Component Modeling and Package Diagram,
12.	Sequence, Collaboration Diagrams and Class Modeling +(Project Architecture
13.	Introduction to J2EE Framework: Components and CORBA Architecture
14.	Software design patterns: Project Demonstrations with Initial Operational Capabilities
15.	Introduction to Software Design Patterns
16.	Principle of Modification of Analysis and Design

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Course Name: Introduction to Modelling and Simulation
Credit Hours: 3 (2-3)

Course Description:
This course is about the modeling and simulation of complex systems by computational methods. The course introduces modeling frameworks for a variety of dynamic systems such as continuous, discrete and hybrid. The course covers the construction of models the analysis of behavioral properties of dynamical systems.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	To acquire knowledge of different types of dynamic systems, their models, properties and analysis method	C2 (Understand)	1,2
CLO-2	To construct models of a variety of complex systems.	C3 (Apply), C6 (Design)	4
CLO-3	To analyze the properties of complex systems by simulation using computational tools.	C4 (Analyze)	3, 5

Course Outline:
Introduction, Model Classification Static and Dynamic Systems, Linear and Nonlinear Systems, Distributed-Parameter Systems, Hybrid and Discrete-Event Systems, Deterministic and Stochastic Systems, Large-Scale Systems, Simulation Languages, Introduction to second-order systems, State-Space Representation, Trajectories and Phase Portraits, The Direction Field, Equilibria, Linear Systems, Linearization of Nonlinear Systems, Periodic Trajectories and Limit Cycles, Existence and Uniqueness of Solution, The Matrix Exponential, The Jordan Canonical Form, Linearization, The Hartman–Grobman Theorem, Singular Perturbations, Exponential Growth and Decay, The Logistic Equation, Models of Epidemics, Predator–Prey System, Introduction to Stability, Lyapunov Stability, Basin of Attraction, The Invariance Principle, Linear Systems and Linearization, Introduction to Feedback Systems, Transfer Functions, Feedback Control, State-Space Models, Minimal Realizations, Pole Placement, State Estimation, The Separation Principle, Optimal Control, Control of Nonlinear Systems, PDE Models, Complex Network, Introduction, Stability of Discrete-Time Systems, Stability of Discrete-Time Linear Systems, Moving-Average Filter, Cobweb Diagrams, Cobweb Diagrams in Economics, The Discrete Logistic Equation, Introduction to System Identification, Least Squares, Regression, Recursive Least Squares, Logistic Regression, Neural Networks, Introduction to System Identification, Least Squares, Regression, Recursive Least Squares, Logistic Regression, Neural Networks, Introduction, Existence and Uniqueness of Solutions, Classification of Linear Second-Order PDEs, The Wave Equation, The D'Alembert Solution, Initial-Value Problem, Separation of Variables, The Heat Equation, Laplace's Equation, Numerical Solution of PDEs, Introduction to Compartmental models, Exponential Growth and Decay, The Logistic Equation, Models of Epidemics, Predator–Prey System, Introduction to Compartmental models, Exponential Growth and Decay, The Logistic Equation, Models of Epidemics, Predator–Prey System, Markov Chains, Regular and Ergodic Markov Chains, Absorbing Markov Chains, Monte Carlo Methods, Random Number Generation, Monte Carlo Integration, Monte Carlo Optimization, Monte Carlo Simulation.

Reference Materials:

1. Introduction to Modelling and Simulation: A Systems Approach by Mark W. Spong (Latest Edition).
2. The Art of Agent-Oriented Modelling (Latest Edition) . The MIT Press, Cambridge, MA., London, England, 2009.

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3. MATLAB: A Practical Introduction to Programming and Problem Solving (Latest Edition) by Attaway Ph.D. Boston University, Stormy

Course Name: Introduction to Modelling and Simulation	
Week #	**Week wise distribution**
1	Introduction, Model Classification Static and Dynamic Systems, Linear and Nonlinear Systems
2	Distributed-Parameter Systems, Hybrid and Discrete-Event Systems, Deterministic and Stochastic Systems, Large-Scale Systems, Simulation Languages
3	Introduction to second-order systems, State-Space Representation, Trajectories and Phase Portraits, The Direction Field, Equilibria
4	Linear Systems, Linearization of Nonlinear Systems, Periodic Trajectories and Limit Cycles
5	Existence and Uniqueness of Solution, The Matrix Exponential, The Jordan Canonical Form, Linearization, The Hartman–Grobman Theorem, Singular Perturbations
6	Exponential Growth and Decay, The Logistic Equation, Models of Epidemics, Predator–Prey System
7	Introduction to Stability, Lyapunov Stability, Basin of Attraction, The Invariance Principle, Linear Systems and Linearization
8	Introduction to Feedback Systems, Transfer Functions, Feedback Control, State-Space Models
9	Minimal Realizations, Pole Placement, State Estimation, The Separation Principle, Optimal Control, Control of Nonlinear Systems, PDE Models, Complex Networks
10	Introduction, Stability of Discrete-Time Systems, Stability of Discrete-Time Linear Systems, Moving-Average Filter, Cobweb Diagrams, Cobweb Diagrams in Economics, The Discrete Logistic Equation
11	Introduction to System Identification, Least Squares, Regression, Recursive Least Squares, Logistic Regression, Neural Networks
12	Introduction, Existence and Uniqueness of Solutions, Classification of Linear Second-Order PDEs, The Wave Equation, The D'Alembert Solution
13	Initial-Value Problem, Separation of Variables, The Heat Equation, Laplace's Equation, Numerical Solution of PDEs
14	Introduction to Compartmental models, Exponential Growth and Decay, The Logistic Equation, Models of Epidemics, Predator–Prey System
15	Markov Chains, Regular and Ergodic Markov Chains, Absorbing Markov Chains, Monte Carlo Methods
16	Random Number Generation, Monte Carlo Integration, Monte Carlo Optimization, Monte Carlo Simulation

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Course Name: Internet of Things
Credit Hours:3 (2-3)

Course Introduction:
The course is designed to teach students the knowledge of network of smart objects and how they are formed and interconnected to build and program such objects that can help improve life standards. They will be able to know various sensors and devices used for this purpose and at the same time program, connect and apply them.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Learn the concept and capabilities of smart thing/object/device identification, and physical principles of sensing	C2 (Understand)	2
CLO-2	Explore IoT enabling technologies, architectures, and standards	C2 (Understand)	2
CLO-3	Understanding of IoT protocol stack, Programming modules and fundamentals of Social IoTs	C4 (Analyze)	3
CLO-4	Apply IoT knowledge to implement small-scale IoT Project	C3 (Apply)	5

Course Outline:
Introduction to IoT, Smart Object Capabilities, IoT Enabling Technologies, IoT Application Domains, – IoT Challenges, IoT Protocol Stack, IoT layered architectures, Basics of Wireless Networking, Basics of Wireless Sensor Networks, IoT Supporting Networks, Introduction to Arduino Programming, Introduction of IoT with Raspberry Pi, – IoT Simulators, Anatomy of IoT Devices, -Smart object concepts, formation and Implementation, Social IoTs

Reference Materials:
- Learning Internet of Things, Peter Waher . (Latest editions).
- IoT: Building Arduino-Based Projects, Peter Waher,Pradeeka Seneviratne, Brian Russell,Drew Van Duren . (Latest editions)
- Mobile Communications, by Jochen Schiller, Addison-Wesley Publishing Company (Latest editions)
- Social Internet of Things: vision, challenges, and trends, Human-centric Computing and Information Sciences, Mozhgan Malekshahi Rad, Amir Masoud Rahmani, Amir Sahafi and Nooruldeen Nasih Qader, (Latest editions)
- Scientific research papers, course slides contain the references

Course Name: Internet of Things	
Week #	**Week wise distribution**
1	– Introduction to IoT, Smart Object Capabilities, IoT Enabling Technologies
2	– IoT Application Domains, Smart Homes, Smart Campus, Smart City, Agricultural Things, Smart Healthcare etc.
3	– IoT Challenges, Interoperability and Standardization, Data Storage Security, Privacy, Fault tolerance, Availability

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4	– IoT Protocol Stack, IoT 3-layered architectures, IoT 5-layered architectures, IoT 6-layered architectures
5	– Basics of Wireless Networking, AD-HOC, MANET – Basics of Wireless Sensor Networks
6	– IoT Supporting Networks, WAN, MAN, CAN, LAN, PAN, BAN, NFC, Nano
7	– Introduction to Arduino Programming, Features of the programming Variables, Loops, Functions etc., Blink project
8	– IoT Simulators
9	– Anatomy of IoT Devices, Understanding PIR motion sensor, programming modules, connectivity, connectivity, its programming and its applications
10	– Anatomy of IoT Devices, Soil Moisture sensor, Implementation and applications, Air Quality sensor, Implementation and applications Ultrasonic sensor, Implementation and applications
11	– Anatomy of IoT Devices, Temperature sensors, IR line follower sensor, Gas sensors, Weekly Mini Project
12	– Anatomy of IoT Devices, Potentiometer, Breadboard, FireSensor, Weekly Mini Project
13	-Smart object concepts and formation
14	-Semester project: Implementation of the smart object
15	– Social IoTs, Vision and Trends, ID and Basic components, Relation Management
16	– Social IoTs, Architecture, SIoTs Platforms, Challenges

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Mathematics and Supporting Courses

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Course Name: Multivariate Calculus
Credit Hours: 3

Objective:
Explain a clear understanding of the fundamental concepts of multivariable calculus and a range of skills allowing the students to work effectively with the concepts like Derivatives as rates of change, computed as a limit of ratios and Integrals as a 'sum,' computed as a limit of Riemann sums.

Learning Outcomes: At the completion of this course, students will be able to learn:

- The fundamental concepts of multivariable calculus
- Range of skills allowing the students to work effectively with the concepts like o Derivatives as rates of change, o Computed as a limit of ratios and o Integrals as a sum, computed as a limit of Riemann sums

Course Outline:
Functions of Several Variables and Partial Differentiation. Multiple Integrals, Line and Surface Integrals. Green‟s and Stoke‟s Theorem. Fourier Series: periodic functions, Functions of any period P-2L, Even and odd functions, Half Range expansions, Fourier Transform; Laplace Transform, Z-Transform.

Reference Materials:

- Multivariable Calculus, 6th edition James, Stewart 2007 Cengage Learning publishers.
- Calculus and Analytical Geometry, 6th edition. Swokowski, Olinick and Pence.1994. Thomson Learning EMEA, Ltd.
- Multivariable Calculus, 5th edition Howard, A. Albert, H. 1995, John Wiley.

Course Name: Multivariate Calculus	
Week #	**Week wise distribution**
1	Functions of Several Variables
2	and Partial Differentiation.
3	Multiple Integrals,
4	Line and Surface Integrals.
5	Green‟s and Stoke‟s Theorem.
6	Fourier Series:
7	periodic functions,
8	Functions of any period P-2L,
9	Even and odd functions,
10	Half Range expansions,
11	Fourier Transform;
12	Fourier Transform;
13	Fourier Transform;
14	Laplace Transform,
15	Z-Transform.
16	Wrap up work and presentations

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Course Name: Linear Algebra and Applications

Credit Hours: 3 (3-0)

Course Introduction:

To provide fundamentals of solution for system of linear equations, operations on system of equations, matrix properties, solutions and study of their properties.

Course Outline:

Algebra of linear transformations and matrices. determinants, rank, systems of equations, vector spaces, orthogonal transformations, linear dependence, linear Independence and bases, eigenvalues and eigenvectors, characteristic equations, Inner product space and quadratic forms

Reference Materials: (or use any other standard and latest books)

- Elementary Linear Algebra by Howard Anton
- Linear Algebra and its Applications by Gibert Strang

Course Name: Linear Algebra and Applications	
Week #	**Week wise distribution**
1	Introduction to Vectors. Solving Linear Equations. Elimination Factorization.
2	Vector Spaces and Subspaces. Orthogonality. DeTerminants. Eigenvalues and Eigenvectors.
3	Linear Transformations. Linear Transformation, Applications of Matrices in Engineering. Graphs and Networks,
4	Markov Matrices, Population, and Economics. Linear Programming. Fourier Series.
5	Linear Algebra for Functions, ,
6	Linear Algebra for Statistics and Probability
7	Computer Graphics
8	Numerical Linear Algebra
9	Complex Vectors and Matrices
10	Discrete Transforms and Simple Applications
11	Cosine Transform
12	The Discrete Fourier Transform
13	Simplification
14	Factorization of the DFT Matrix
15	Fast Fourier Transforms. The Discrete Time Fourier Transform. The Z-Transform
16	Wrap up work and presentations

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Course Name: Probability and Statistics
Credit Hours: 3 (3-0)

Objectives:
To enable the students to think and use the applications of calculus and analytical geometry. Map other concepts with computer science perspectives. To understand the fundamental concepts of statistics and probability. The role of probability in evaluation in computer science in related domain research.

Learning Outcome:
- Use statistical vocabulary, Construct various frequency distributions of grouped and ungrouped data.
- Calculate and interpret descriptive statistics of samples and populations. (Measures of central tendency, measures of dispersion.)
- Calculate simple probabilities, Find the mean and variance of a probability distribution including the binomial distribution.
- Understand and calculate expected values, Calculate the probabilities or scores of normal distributions and the normal approximation of the binomial distribution.
- Use the Central Limit Theorem to calculate the probabilities of the mean for any distribution.
- Formulate, calculate and interpret hypotheses test for one parameter and to compare two parameters, for both large and Small samples, Z and T for one two samples.

Course Outline:
Introduction to Statistics and Data Analysis, Statistical Inference, Samples, Populations, and the Role of Probability. Sampling Procedures. Discrete and Continuous Data. Statistical Modeling. Types of Statistical Studies. Probability: Sample Space, Events, Counting Sample Points, Probability of an Event, Additive Rules, Conditional Probability, Independence, and the Product Rule, Bayes‟ Rule. Random Variables and Probability Distributions. Mathematical Expectation: Mean of a Random Variable, Variance and Covariance of Random Variables, Means and Variances of Linear Combinations of Random Variables, Chebyshev‟s Theorem. Discrete Probability Distributions. Continuous Probability Distributions. Fundamental Sampling Distributions and Data Descriptions: Random Sampling, Sampling Distributions, Sampling Distribution of Means and the Central Limit Theorem. Sampling Distribution of S2, t -Distribution, FQuantile and Probability Plots. Single Sample and One- and Two-Sample Estimation Problems. Single Sample and One- and Two-Sample Tests of Hypotheses. The Use of PValues for Decision Making in Testing Hypotheses Single Sample and One - and TwoSample Tests), Linear Regression and Correlation. Least Squares and the Fitted Model, Multiple Linear Regression and Certain, Nonlinear Regression Models, Linear Regression Model Using Matrices, Properties of the Least Squares Estimators.

Reference Materials:

- Probability and Statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying E. Ye, Pearson; 9th Edition January 6, 2011). ISBN -10: 0321629116
- Probability and Statistics for Engineers and Scientists by Anthony J. Hayter, Duxbury Press; 3rd Edition (February 3, 2006), ISBN-10: 0495107573
- Schaum's Outline of Probability and Statistics, by John Schiller, R. Alu Srinivasan and Murray Spiegel, McGraw-Hill; 3rd Edition 2008). ISBN -10: 0071544259

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Course Name: Probability and Statistics	
Week #	**Week wise distribution**
1	Introduction to Statistics and Data Analysis, Statistical Inference, Samples,
2	Populations, and the Role of Probability. Sampling Procedures. Discrete and Continuous Data. Statistical Modeling. Types of Statistical Studies.
3	Probability: Sample Space, Events, Counting Sample Points, Probability of an Event,
4	Additive Rules, Conditional Probability, Independence, and the Product Rule, Bayes‟ Rule.
5	Random Variables and Probability Distributions. Estimation Problems.
6	Single Sample and One- and Two-Sample Tests of Hypotheses. The Use of PValues for Decision Making in
7	Mathematical Expectation: Mean of a Random Variable, Variance and Covariance of Random Variables, Means and Variances of Linear Combinations of Random Variables
8	Chebyshev‟s Theorem. Discrete Probability Distributions. Continuous Probability Distributions. Fundamental Sampling Distributions and Data Descriptions: Random Sampling, Sampling Distributions,
9	Sampling Distribution of Means and the Central Limit Theorem.
10	Sampling Distribution of S2, t -Distribution,
11	F Quantile and Probability Plots.
12	, Single Sample and One- and Two-Sample
13	Testing Hypotheses Single Sample and One - and Two Sample Tests), Linear Regression and Correlation.
14	Least Squares and the Fitted Model, Multiple Linear Regression and Certain,
15	Nonlinear Regression Models, Linear Regression Model Using Matrices, Properties of the Least Squares Estimators.
16	Wrap up work and presentations

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Course Name: Technical and Business Writing
Credit Hours: 3 (3-0)

Course Introduction:
Students in the senior level needs good technical writing skills not only for writing project report but also useful for them to communicate their resume and get place in the market. This is a high level course which provide useful knowledge to the students for writing proposals etc. Further, the course aims at augmenting students' proficiency in technical writing in order to sensitize them to the dynamics, challenges, and needs of the modern world characterized by technologically advanced social, cultural, and corporate settings. It will focus on students' ability to effectively convey and exchange information. in cross-cultural, international, and multinational milieu necessitated by the emergence of global society.

Course Outline:
Overview of technical reporting, use of library and information gathering, administering questionnaires, reviewing the gathered information; Technical exposition; topical arrangement, exemplification, definition, classification and division, casual analysis, effective exposition, technical narration, description and argumentation, persuasive strategy, Organizing information and generation solution: brainstorming, organizing material, construction of the formal outline, outlining conventions, electronic communication, generation solutions. Polishing style: paragraphs, listening sentence structure, clarity, length and order, pomposity, empty words, pompous vocabulary, document design: document structure, preamble, summaries, abstracts, table of contents, footnotes, glossaries, cross- referencing, plagiarism, citation and bibliography, glossaries, index, appendices, typesetting systems, creating the professional report; elements, mechanical elements and graphical elements. Reports: Proposals, progress reports, Leaflets, brochures, handbooks, magazines articles, research papers, feasibility reports, project reports, technical research reports, manuals and documentation, thesis. Electronic documents, Linear verses hierarchical structure documents.

Reference Materials: (or use any other standard and latest books)
- Technical Report Writing, by Pauley and Riordan, Houghton Mifflin Company, 8th Edition.
- Effective Technical Communication by Ashraf Rizvi, Tata McGraw-Hill.

Course Name: Technical and Business Writing	
Week #	**Week wise distribution**
1	Characteristics of Academic, Public, Work and Electronic Communities. Myths and Realities about Writing.
2	Effective Writing: Discovering and Planning; Purpose, Thesis, and Audience; Drafting: Drafting Collaboratively, Drafting in Digital Environments;
3	Revising, Editing, and Proofreading. Paragraphs: Unfocused Paragraphs, Incoherent Paragraphs, Poorly Developed Paragraphs,
4	Special-Purpose Paragraphs. Unclear, Clear and Emphatic Sentences. Reasoning Critically. Reading Critically. Arguing Persuasively and Logically.
5	Designing Documents. Writing in Online Communities. Speaking Effectively.
6	Academic Writing for Social and Natural Sciences: Goals of Writing, Audiences,
7	Writing Tasks, Types of Writing: Abstract, Informative Report, Lab Report,
8	Research Report,
9	Project Reports Public Writing:
10	Goals of Public Writing, Public Audiences,
11	Public Writing Tasks, Types of Public Writing,
12	Public Flyer, Letter to the Editor.
13	Researching and Writing: Types of Research Writing,

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14	Developing a Research Question, Developing a Preliminary Thesis, Creating a Research File and a Timeline,
15	Reading and Note taking, Summarizing, Paraphrasing, and Synthesizing. Writing a Position Paper.
16	Wrap up work and presentations

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Elective Supporting Courses

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General Education Requirement

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Course Name: Introduction to Information and Communication Technologies

Credit Hours: 3 (2-3)

Course Description:

This is an introductory course on Information and Communication Technologies. The course is designed for students with little or no computer experience. The course introduces ICT and computer concepts, hardware components, basic computer operations and use of software applications to solve problems. Students will have complete understanding and use of ICT technologies, personal computers, system, and application software such as Window, Microsoft Office, Internet, WWW, Email and Social Media applications.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand basics of computing technology	C1 (Knowledge)	2
CLO-2	Do number systems conversions and arithmetic	C2 (Understand)	4
CLO-3	Have knowledge of types of software	C2(Understand)	2
CLO-4	Have knowledge of computing related technologies	C3 (Apply)	2

Course Outlines:

- Introduction to Computers and ICT
- Data and Information
- Applications of Computer and ICT
- Structure of Computer System (Input and Output Unit, CPU and Memory)
- Types of Computers
- Input devices and Output Devices
- Computer Storage devices (Primary and Secondary Storage)
- Computer software (application and system)
- Operating System and Its Basic Functions
- Introduction to networking, network elements and Types
- Introduction to Database, Characteristics of Relational Database
- Internet applications; www, email.
- Computer Security Threats, Viruses, Worms.
- Number Systems and their Inter-conversion
- Practical Work: Basics of window operating system, MS Word, MS Excell, MS Power Point, MS Access, Internet, WWW, Email, social media.

Reference Materials:

- Discovering Computers: Your Interactive Guide to the Digital World by Gary BS and Misty EV, Latest Edition.
- Introduction to Computers" by Peter Norton, Latest Edition (McGraw Hill)
- Using Information Technology: A Practical Introduction to Computer &Communications" by Williams Sawyer, Latest Edition (McGraw Hill).
- Fundamentals of Information Technology, by Alexis Leon and Mathews Leon, Latest Edition.

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Introduction to Information and Communication Technologies	
Week #	**Week wise distribution**
1	Introduction to Computers and ICT , Data and Information , Applications of Computer and ICT
2	Structure of Computer System (Input and Output Unit, CPU and Memory), Types of Computers
3	Input devices and Output Devices
4	Computer software (application and system)
5	Operating System and Its Basic Functions
6	Introduction to networking, network elements and Types
7	Introduction to Database, Characteristics of Relational Database
8	Internet applications; www, email.
9	Computer Security Threats, Viruses, Worms.
10	Number Systems and their Inter-conversion
11	Practical Work: Basics of window operating system
12	Practical Work: MS Word
13	Practical Work: MS Excell
14	Practical Work: MS Power Point
15	Practical Work: MS Access
16	Practical Work: Internet, WWW, Email, social media.

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Course Name: Discrete Structures
Credit Hours: 3 (3-0)

Course Introduction:
Introduces the foundations of discrete mathematics as they apply to Computer Science, focusing on providing a solid theoretical foundation for further work. Further, this course aims to develop understanding and appreciation of the finite nature inherent in most Computer Science problems and structures through study of combinatorial reasoning, abstract algebra, iterative procedures, predicate calculus, tree and graph structures. In this course more emphasis shall be given to statistical and probabilistic formulation with respect to computing aspects.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Understand the key concepts of Discrete Structures such as Sets, Permutations, Relations, Graphs and Trees etc.	C1 (Knowledge)	2
CLO-2	Apply formal logic proofs and/or informal, but rigorous, logical reasoning to real problems, such as predicting the behavior of software or solving problems such as puzzles	C2(Apply)	4
CLO-3	Apply discrete structures into other computing problems such as formal specification, verification, databases, artificial intelligence, and cryptography	C2(Apply)	4
CLO-4	Differentiate various discrete structures and their relevance within the context of computer science, in the areas of data structures and algorithms, in particular	C3(Differentiate)	3

Course Outlines:
Mathematical reasoning, propositional and predicate logic, rules of inference, proof by induction, proof by contraposition, proof by contradiction, proof by implication, set theory, relations, equivalence relations and partitions, partial orderings, recurrence relations, functions, mappings, function composition, inverse functions, recursive functions, Number Theory, sequences, series, counting, inclusion and exclusion principle, pigeonhole principle, permutations and combinations. Algorithms, Searching and Sorting Algorithms, elements of graph theory, planar graphs, graph coloring, Graph Algorithms, euler graph, Hamiltonian path, rooted trees, traversals.

Reference Materials: (or use any other standard and latest books)
- Discrete Mathematics and Its Applications, Latest edition by Kenneth H. Rosen
- Discrete Mathematics with Applications, Latest Edition by Susanna S. Epp
- Discrete Mathematics, Latest edition by Richard Johnson Baugh
- Discrete Mathematical Structures, Latest edition by Kolman, Busby and Ross
- Discrete and Combinatorial Mathematics: An Applied Introduction by Ralph P. Grimaldi
- Logic and Discrete Mathematics: A Computer Science Perspective by Winifred Grassman

Discrete Structures	
Week #	**Week wise distribution**
1	Mathematical reasoning
2	Mathematical reasoning
3	introduction to logic

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4	propositional and
5	predicate calculus; negation disjunction
6	conjunction; implication and equivalence
7	truth tables
8	predicates; quantifiers
9	natural deduction; rules of Inference; methods of proofs
10	use in program proving; resolution principle
11	Set theory: Paradoxes in set theory; inductive definition of sets and proof by induction; Relations
12	representation of relations by graphs; properties of relations, equivalence relations and partitions; Partial function theory
13	Elementary combinatorics; counting techniques
14	recurrence relation; generating functions
15	Graph Theory: elements of graph theory, Planar Graphs, Graph Colouring
16	Wrap up Euler graph, Hamiltonian path, trees and their applications

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Course Name: Calculus and Analytical Geometry
Credit Hours: 3 (3-0)

Course Introduction:
To provide foundation and basic ground for calculus and analytical geometry background.

Course Outline:
Limits and Continuity; Introduction to functions, Introduction to limits, Techniques of funding limits, Indeterminate forms of limits, Continuous and discontinuous functions and their applications, Differential calculus; Concept and idea of differentiation, Geometrical and Physical meaning of derivatives, Rules of differentiation, Techniques of differentiation, Rates of change, Tangents and Normals lines, Chain rule, implicit differentiation, linear approximation, Applications of differentiation; Extreme value functions, Mean value theorems, Maxima and Minima of a function for single-variable, Concavity, Integral calculus; Concept and idea of Integration, Indefinite Integrals, Techniques of integration, Riemann sums and Definite Integrals, Applications of definite integrals, Improper integral, Applications of Integration; Area under the curve, Analytical Geometry; Straight lines in R3, Equations for planes.

Reference Materials:
- Calculus and Analytic Geometry by Kenneth W. Thomas.
- Calculus by Stewart, James.
- Calculus by Earl William Swokowski; Michael Olinick; Dennis Pence; Jeffery A. Cole

Course Name: Quantitative Reasoning – 2 (Calculus and Analytical Geometry)	
Week #	**Week wise distribution**
1	Real Numbers and the Real Line, Coordinates, Lines, and Increments, Functions, Shifting Graphs, Trigonometric Functions.
2	Limits and Continuity: Rates of Change and Limits, Rules for Finding Limits,
3	Target Values and Formal Definitions of Limits, Extensions of the Limit Concept,
4	Continuity, Tangent Lines. Derivatives
5	The Derivative of a Function,
6	Differentiation Rules, Rates of Change,
7	Derivatives of Trigonometric Functions,
8	The Chain Rule, Implicit Differentiation and Rational Exponents.
9	Applications of Derivatives. Integration: Indefinite Integrals, Integration by Substitution,
10	Natural Logarithms, The Exponential Function, ax and logax, Growth and Decay, L'Hôpital's Rule, Relative Rates of Growth,
11	Inverse Trigonomic Functions, Derivatives of Inverse Trigonometric Functions; Hyperbolic Functions.
12	Conic Sections, Parametrized Curves, and Polar Coordinates. Graphing in Polar Coordinates. Polar Equations for Conic Sections.
13	Integration in Polar Coordinates. Vectors and Analytic Geometry in Space; Vectors in the Plane Dot Products; Vector-Valued Function Cartesian Rectangular) Coordinates and Vectors in Space.
14	Definite Integrals, Substitution in Definite Integrals. Numerical Integration. Applications of Integrals. Transcendental Functions: Inverse Functions and Their Derivatives,

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15	Dot Products; Cross Products. Lines and Planes in Space; Cylinders and Quadric Surfaces; Cylindrical and Spherical Coordinates.
16	Wrap up work and presentations

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Course Name: Professional Practices
Credit Hours: 2 (2-0)

Course Introduction:
A Computing graduate as professional has some responsibilities with respect to the society. This course develops student understanding about historical, social, economic, ethical, and professional issues related to the discipline of Computing. It identifies key sources for information and opinion about professionalism and ethics. Students analyze, evaluate, and assess ethical and professional computing case studies.

CLO No.	Course Learning Outcomes	Bloom's Taxonomy	PLOs
CLO-1	Define the key terms related to Computing, including Software Engineering, Information Technology, and Computer Science	C1(Knowledge)	2
CLO-2	Explain the role of professional societies, ethics, and life-long learning in computing careers	C2(Understand)	2
CLO-3	Apply basic legal principles, such as intellectual property rights and cyber law, to software-related scenarios	C3(Apply)	3,4

Course Outline:
Historical, social, and economic context of Computing (software engineering, Computer Science, Information Technology); Definitions of Computing (software engineering, Computer Science, Information Technology) subject areas and professional activities; professional societies; professional ethics; professional competency and life-long learning; uses, misuses, and risks of software; information security and privacy; business practices and the economics of software; intellectual property and software law (cyber law); social responsibilities, software related contracts, Software house organization. Intellectual Property Rights, The Framework of Employee Relations Law and Changing Management Practices, Human Resource Management and IT, Health and Safety at Work, Software Liability, Liability and Practice, Computer Misuse and the Criminal Law, Regulation and Control of Personal Information. Overview of the British Computer Society Code of Conduct, IEEE Code of Ethics, ACM Code of Ethics and Professional Conduct, ACM/IEEE Software Engineering Code of Ethics and Professional Practice. Accountability and Auditing, Social Application of Ethics.

Reference Materials: (or use any other standard and latest books)

- Professional Issues in Software Engineering by Frank Bott, Allison Coleman, Jack Eaton and Diane Rowland, CRC Press; Latest Edition.
- Computer Ethics by Deborah G. Johnson, Pearson; Latest Edition
- A Gift of Fire: Social, Legal, and Ethical Issues for Computing and the Internet by Sara Baase, Prentice Hall; Latest Edition.
- Applied Professional Ethics by Gregory R. Beabout, University Press of America Latest Edition.

Course Name: Financial Accounting	
Week #	**Week wise distribution**
1	Computing Profession, Computing Ethics,
2	Philosophy of Ethics. The Structure of Organizations,

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3	Finance and Accounting, Anatomy of a Software House, Computer Contracts,
4	Intellectual Property Rights,
5	The Framework of Employee Relations Law and Changing Management Practices,
6	Human Resource Management and IT,
7	Health and Safety at Work,
8	Software Liability,
9	Liability and Practice,
10	Computer Misuse and the Criminal Law,
11	Regulation and Control of Personal Information.
12	Overview of the British Computer Society Code of Conduct,
13	IEEE Code of Ethics,
14	ACM Code of Ethics and Professional Conduct, ACM/IEEE
15	Software Engineering Code of Ethics and Professional Practice. Accountability and Auditing, Social Application of Ethics.
16	Wrap up work and presentations

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Deficiency Courses

Course Name: Mathematics I

Credit Hours: 3 (3-0)

Course Outline:

Complex Numbers, Matrices and Determinants, Vectors, Sequences, Miscellaneous Series, Permutation, Combination, Probability, Mathematical Induction and Binomial Theorem, Functions and Graphs, Linear Programming, Trigonometric Identities, Application of Trigonometry, Graph of Trigonometric and Inverse Trigonometric Functions and Solutions of Trigonometric Equations

Reference Materials:

A Textbook of Mathematics for Grade XI by Prof Dr. Gulzar Ali Khan and Prof Dr. Islam Noor

Course Name: Mathematics II

Credit Hours: 3 (3-0)

Course Outline:

Functions and Limits, Differentiation, Higher Order Derivative and Applications, Differentiation of Vector Functions, Integrations, Plane Analytic Geometry, Conics, Differential Equations, Partial Differentiation, Numerical Methods

Reference Materials:

A Textbook of Mathematics for Grade XII by Prof. Mumtaz Khan

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Annexure A3

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NEW AND REVISED COURSES OF M.PHIL AND PH.D PROGRAMS
(Spring 2025 onwards)



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Course Name: Modelling and Simulation
Type: Core course
Course Code: CS811
Credit Hours: 3

Course Outline:

Introduction: Systems Engineering, Model Classification Static and Dynamic Systems, Linear and Nonlinear Systems, Distributed-Parameter Systems, Hybrid and Discrete-Event Systems, Deterministic and Stochastic Systems, Large-Scale Systems, Simulation Languages
Second-Order Systems: Introduction, State-Space Representation, Trajectories and Phase Portraits, The Direction Field, Equilibria, Linear Systems, Linearization of Nonlinear Systems, Periodic Trajectories and Limit Cycles
System Fundamentals: Introduction, Existence and Uniqueness of Solution, The Matrix Exponential, The Jordan Canonical Form, Linearization, The Hartman–Grobman Theorem, Singular Perturbations
Compartmental Models: Introduction, Exponential Growth and Decay, The Logistic Equation, Models of Epidemics, Predator–Prey System
Stability: Introduction, Lyapunov Stability, Basin of Attraction, The Invariance Principle, Linear Systems and Linearization
Feedback Systems: Transfer Functions, Feedback Control, State-Space Models, Minimal Realizations, Pole Placement, State Estimation, The Separation Principle, Optimal Control, Control of Nonlinear Systems, PDE Models, Complex Networks
Partial Differential Equation Models: Introduction, Existence and Uniqueness of Solutions, Classification of Linear Second-Order PDEs, The Wave Equation, The D'Alembert Solution, Initial-Value Problem, Separation of Variables, The Heat Equation, Laplace's Equation, Numerical Solution of PDEs
Complex Networks: Introduction, Examples of Complex Networks, Graph Isomorphism, Connectivity, Trees, Bipartite Graphs, Planar Graphs, Graphs and Matrices, Matlab Graph Functions, Network Metrics, Degree Distribution, Centrality, Clustering, Random Graphs, Synchronization in Networks

Reference Materials:

1. Introduction to Modelling and Simulation: A Systems Approach by Mark W. Spong (Latest Edition).
2. The Art of Agent-Oriented Modelling (Latest Edition) . The MIT Press, Cambridge, MA., London, England, 2009.
3. MATLAB: A Practical Introduction to Programming and Problem Solving (Latest Edition) by Attaway Ph.D. Boston University, Stormy

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Course Name: Computational Methods in Modelling and Simulation
Type: Core course
Course Code: CS812
Credit Hours: 3

Course Outline:

Introduction: Model Classification Static and Dynamic Systems Distributed-Parameter Systems, Hybrid and Discrete-Event Systems, Deterministic and Stochastic Systems, Large-Scale Systems, Simulation Languages.
Compartmental Models: Introduction, Exponential Growth and Decay, The Logistic Equation, Models of Epidemics, Predator–Prey System
Discrete-Time Systems: Introduction, Stability of Discrete-Time Systems, Stability of Discrete-Time Linear Systems, Moving-Average Filter, Cobweb Diagrams, Cobweb Diagrams in Economics, The Discrete Logistic Equation
Numerical Methods: Introduction, Numerical Differentiation, Numerical Integration, Numerical Solution of ODEs, Euler Predictor–Corrector Method, Runge–Kutta Methods, Stiff System, Event Detection, Simulink
Optimization: Introduction, Unconstrained Optimization, Iterative Search, Gradient Descent, Newton's Method, Case Study: Numerical Inverse Kinematics, Constrained Optimization, Equality Constraints, Inequality Constraints, Convex Optimization
System Identification: Introduction, Least Squares, Regression, Recursive Least Squares, Logistic Regression, Neural Networks
Stochastic Systems: Markov Chains, Regular and Ergodic Markov Chains, Absorbing Markov Chains, Monte Carlo Methods, Random Number Generation, Monte Carlo Integration, Monte Carlo Optimization, Monte Carlo Simulation

Reference Materials:

1. Introduction to Modelling and Simulation: A Systems Approach by Mark W. Spong (Latest Edition)
2. The Art of Agent-Oriented Modelling (Latest Edition) . The MIT Press, Cambridge, MA., London, England, 2009.
3. MATLAB: A Practical Introduction to Programming and Problem Solving (Latest Edition) by Attaway Ph.D. Boston University, Stormy

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Course Name: Quantum Computing
Type: Core course
Course Code: CS815
Credit Hours: 3

Course Outline:

Introduction: Evolution of Classical and Quantum Computers, Comparison of Classical and Quantum Computers, Quantum Supremacy, Qubit Introduction
Mathematical Foundations: Probability, Complex Numbers, Matrix, Matrix Operations, Linear Transformation, Probability of Measurement, Orthonormality, Orthonormality in Bracket Notation, Introduction to Basis Vectors, Basis Vectors for Spin, Introduction to Degrees of Freedom, Degrees of Freedom for a Single Qubit, Global & Relative Phase, Bloch Sphere, Tensor Products, Tensor Products of Matrices, Multi-Qubit Basis Vectors, Brackets of Tensor Terms, Multi-Qubit Tensor Algebra, Entanglement & Degrees of Freedom, Entanglement in Bracket Notation, Change of Basis, Multi-Qubit Measurements, Partial Measurement of Multiple Qubits, Square Matrices in Bracket Notation, Cryptography with Entanglement, Expected Value of Measurables, Expected Value in Matrix Form, Deconstruction of Hermitians, Deconstruction of Unitary Matrices, Proof of No-Cloning Theorem, Superdense Coding, Quantum Teleportation, Bell's Theorems
Gates & Circuits: Phase, Phase and Bloch Sphere, Classical Gates, IBM Signup, Quantum Gates, Quantum Circuits
Teleportation: Introduction to Teleportation, Superdense Coding, Quantum Teleportation, Teleportation in Qiskit
Quantum Algorithms: Introduction to Bernstein Vazirani, Bernstein Vazirani Algorithm, Improving Codes, Introduction to Deutsch, Deutsch Algorithm, Creating Algorithm, Deutsch in Qiskit, Introduction to Grover's, Classical Search Algorithm, Applying Grover's on Qiskit, Introduction to Shor's, Shor's Algorithm, Quantum Fourier Transform, Quantum Phase Estimation

Reference Materials:

1. Quantum Computing for Computer Scientists by Noson S. Yanofsky , Mirco A. Mannucci (Latest Edition)
2. Quantum Computation and Quantum Information by Michael A. Nielsen (Author), Isaac L. Chuang (Latest Edition)

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Course Name: Advanced Artificial Intelligence
Type: Core course
Course Code: CS857
Credit Hours: 3

Course Outline:

Introduction to Deep Learning, Artificial Neural Networks: Perceptron and Multi-Layer Perceptron, Understand structure and limitations of single layer perceptron, explain the architecture and functionality of multi-layer perceptron (MLPs), Artificial Neural Networks: Backpropagation, Optimisation, and Implementation, Discover activation functions, Implement MLP using backpropagation, Training Neural Networks: Hyperparameter Selection, Optimization Algorithms, Select various hyperparameters to avoid overfitting. Understand the working of different optimisation algorithms, Convolutional Neural Networks, Components of CNN, Dilated Convolution, Transposed Convolution, Common CNN Architectures, VGG, Inception, Residual Networks, Dense Networks (Transfer Learning), Computer Vision - I: Object Detection, Computer Vision - II: Semantic Segmentation, Natural Language Processing-I, Sequence Modelling using RNN, GRU, LSTM, Machine Translation using seq2seq models and Cross Attention, Transformers and self-attention, Autoencoders, VAEs, GANs, Diffusion Models, Introduction to Reinforcement Learning and Q-Learning, epsilon-greedy policy, Identifying and mitigating biases in data and deep models.

Reference Material:

1. Textbook: Deep Learning by Ian Goodfellow, Yoshua Benjio, and Aaron Courville
2. Any material (papers, notes, slides, video) on the web from credible sources.

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Course Title: Principles of Model Checking
Course Code: CS841
Credit Hours: 3

Course Outline:

Introduction, Automata, Synchronized Product, Synchronization by Message Passing, Synchronization by Shared Variables, Temporal Logic, The Formal Syntax of Temporal Logic , The Semantics of Temporal Logic, PLTL and CTL: Two Temporal Logics, The Expressivity of CTL * , Model Checking, CTL Model Checking, PLTL Model Checking, The State Explosion Problem , Symbolic Model Checking, Symbolic Computation of State Sets, Binary Decision Diagrams (BDD), Representing Automata by BDDs, BDD-based Model Checking, Reachability Properties, Reachability in Temporal Logic, Safety Properties, Safety Properties in Temporal Logic, Safety Properties in Practice, Liveness Properties, Liveness in Temporal Logic, Bounded Liveness, Deadlock-freeness, Deadlock-freeness in Temporal Logic, Fairness Fairness in Temporal Logic, Fairness and Nondeterminism, Fairness Properties and Fairness Hypotheses, Strong Fairness and Weak Fairness, Abstraction Methods, Model Checking tools: SMV, SPIN, UPPAAL

Reference Material:

1. B. Bérard et al.: Systems and Software Verification: Model-Checking Techniques and Tools, Springer, Latest Edition.
2. C. Baier and J.-P. Katoen: Principles of Model Checking, MIT Press, Latest Edition

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Course Title: Verification of Hybrid and Real-Time Systems
Course Code: CS844
Credit Hours: 3

Course Outline:
Introduction to Real-Time and Hybrid Systems, Automata, Clocks, Invariants, Guards, Timed-Automata, Semantics of Timed Automata, Network of Time-Automata, Composition of Network of Timed-Automata, Semantics of Composition of Network of Timed-Automata, Modelling examples: Rail-Road Crossing; Gas Burner, Timed Computation Tree Logic (TCTL), Regions, Zones, Reachability Graph, Modelling in UPPAAL Model Checker, Verification of Properties in UPPAAL, Hybrid Automata, Composition of Hybrid Automata, HyTech model checker, Case studies.

Reference Material:

1. C. Baier and J.-P. Katoen: Principles of Model Checking, MIT Press, Latest Edition
2. Raskin, JF. (2005). An Introduction to Hybrid Automata. In: Hristu-Varsakelis, D., Levine, W.S. (eds) Handbook of Networked and Embedded Control Systems. Control Engineering.
3. Henzinger, T.A., Ho, PH., Wong-Toi, H. (1997). HyTech: A model checker for hybrid systems. In: Grumberg, O. (eds) Computer Aided Verification. CAV 1997. Lecture Notes in Computer Science, vol 1254. Springer, Berlin, Heidelberg. https://doi.org/10.1007/3-540-63166-6_48

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Course Name: Petri Nets
Course Code: CS845
Credit Hours: 3

Course Outline:

Introduction to graph, Bipartite graphs, Introduction to Petri Nets ,Properties, Marking graph, Mathematical representation of Petri Nets, P-Invariants, T-Invariants, Liveness, Conflicts, Matrix representation of Petri Nets, Algorithm for P/T-Invariants, Reachability, Non-Autonomous Petri Nets, Synchronized Petri Nets ,Timed Petri Nets, Stochastic Petri Nets, Continuous and Hybrid Petri Nets, Timed Continuous Petri Nets, Timed Hybrid Petri Nets, Examples of Industrial Applications, Snoopy Tool, Model construction in Snoopy, Model Validation in Snoopy, Simulation in Snoopy, Properties Verification in Charlie tool

Reference Material:

1. Discrete, Continuous, and Hybrid Petri Nets By René David, Hassane Alla (Latest Edition)
2. Coloured Petri Nets: Modelling and Validation of Concurrent Systems by Kurt Jensen (Author), Lars M. Kristensen (Latest Edition)

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Course Name: Colored Petri Nets
Course Code: CS846
Credit Hours: 3

Course Outline:

Modelling and System Development, Formal Modelling and Verification, Non-hierarchical Coloured Petri Nets Structure, Enabling and Occurrence of Transitions, Concurrency and Conflict, Guards, Interactive and Automatic Simulation, Functional Programming, Colour Sets, Expressions and Types, Functions, Recursion and Lists, Patterns, Multisets, Net Structure and Inscriptions, Enabling and Occurrence of Steps, Hierarchical Coloured Petri Nets, Modules and Interfaces, Module Instances and Hierarchy, Instance Folding and Module Parameterisation, Model Parameterisation, Fusion Sets, Unfolding Hierarchical CPN Models, Formal Definition of Hierarchical Coloured Petri Nets, Module Composition, Instances and Compound Places, Enabling and Occurrence of Steps, Protocol for State Space Analysis, State Spaces, Strongly-Connected-Component Graphs, Behavioural Properties, Error Diagnostics and Counterexamples, State Space Reduction Methods, Reachability Properties, Boundedness Properties, Home Properties, Liveness Properties, Fairness Properties, Timed Coloured Petri Nets, Timed multisets, Net Structure, Enabling and Occurrence of Steps, Timed Protocol for Performance Analysis, Data Collection from the Occurring Binding Elements, Data Collection from the Markings Reached,, Collecting Data from the Final, Message Sequence Charts, Examples of Industrial Applications

Reference Material:

1. Coloured Petri Nets: Modelling and Validation of Concurrent Systems
2. by Kurt Jensen (Author), Lars M. Kristensen (Latest Edition)
3. Discrete, Continuous, and Hybrid Petri Nets By René David, Hassane Alla (Latest Edition)

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Course Name: Applications of AI in Software Engineering
Course code: CS858
Credit Hours: 3

Course Contents:

AI-driven software development life cycle, AI for advanced software engineering, AI techniques for software engineering, Applications of ChatGPT in software engineering, Enhancing software development efficiency through AI-powered code generation, AI-powered debugging, Innovations in AI and their impact on software engineering, Problem Solving by Searching (Informed searching, Uninformed searching, Local searching); Constraint Satisfaction Problems; Adversarial Search (Min-max algorithm, Alpha beta pruning, Game-playing); Learning (Unsupervised learning, Supervised learning, Reinforcement learning); Latest trends in the field of AI-driven software engineering (Google's data center efficiency optimization, Microsoft's visual studio code completion, GitHub's CodeQL for security analysis, OpenAI's Codex for code generation, IBM's Watson for software engineering, Explainable AI (XAI), Integration of AI with DevOps practices, AI for code generation and AutoML), Generative AI, Superintelligence, Ethical considerations.

Reference Material:

1. Hopgood, Adrian A. *Intelligent systems for engineers and scientists: a practical guide to artificial intelligence*. Latest Edition.
2. Black, R., Davenport, J., Olszewska, J., Rößler, J., Smith, A. L., & Wright, J. (2022). *Artificial Intelligence and Software Testing: Building systems you can trust*. BCS Press, 2022.
3. Marar, H. W. (2024). Advancements in software engineering using AI. *Computer Software and Media Applications*, *6*(1), 3906, 2024.
4. Russell, S. J., & Norvig, P. (2016). *Artificial intelligence: a modern approach*. Pearson, 2016
5. Oroy, K., & Liu, C. (2024). *AI Techniques for Software Engineering: Current Trends and Future Prospects* (No. 12077). EasyChair, 2024.
6. Neelofar, N., & Aleti, A. (2024, February). Towards Reliable AI: Adequacy Metrics for Ensuring the Quality of System-level Testing of Autonomous Vehicles. In *Proceedings of the 46th IEEE/ACM International Conference on Software Engineering* (pp. 1-12), 2024.

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Course Name: Software Process Improvement
Course Code: CS859
Credit Hours: 3

Course Contents:

Software Process Improvement (SPI), benefits and standard SPI models like Capability Maturity Model Integration (CMMI), etc. SPI initiatives in various phases of Software Development Life Cycle (SDLC). Rapid process improvement through advanced AI techniques. Impact of SPI on software quality, critical success factors for SPI implementation, Setting-up a CMMI Initiative (Approach to Continuous Improvement, CMMI Improvement Structure and Teams, Planning the Improvement Cycle, Implementation of Improvements). CMMI Level-2 Implementation (Project Management, Supplier Agreement Management, Requirements Development and Management, Configuration Management, Process and Product Quality Assurance, Measurement and Analysis). CMMI Level-3 Implementation (Organization Process Focus, Organization Process Definition, Organization Training, Requirements, Design, and Development, Verification and Validation, Integrated Project Management, Risk Management, Decision Analysis and Resolution). CMMI Level 4 and 5 Implementation (Organization Process Performance, Quantitative Project Management, Organization Innovation and Deployment). Project Management Maturity, Comparison of Project Management Maturity Models, such as PMMM, CMMI, OPM3 etc. SPI in the context of Global Software Engineering (Various GSD models such as Software Outsourcing Vendors'Readiness Model (SOVRM), Outsourcing Contract Management Model, Software Outsourcing Partnership, Complex outsourcing Relationships management model etc.)

Reference Material:

1. Buede, D. M., & Miller, W. D. (2024). *The engineering design of systems: models and methods*. John Wiley & Sons, 2024.
2. O'Regan, G. (2010). *Introduction to software process improvement*. Springer Science & Business Media, 2010.
3. Niazi, M., Wilson, D., & Zowghi, D. (2005). A framework for assisting the design of effective software process improvement implementation strategies. *Journal of systems and software*, *78*(2), 204-222, 2005.
4. Humayun, M., Niazi, M., Jhanjhi, N. Z., Mahmood, S., & Alshayeb, M. (2023). Toward a readiness model for secure software coding. *Software: Practice and Experience*, *53*(4), 1013-1035, 2023.
5. Abasova, G. (2024). *Organizational challenges with growing project management maturity* (Doctoral dissertation, Vilniaus universitetas.), 2024.

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Course Name: Advanced Theory of Computation
Course Code: CS860
Credit Hours: 3

Course Outline:

Finite Automata: DFA, NFA, ε-NFA, equivalence among automata, closure properties, minimization of finite automata
Regular Expressions: Kleene's theorem, applications of regular expressions in real-world systems (e.g., search engines, text processing)
Determinism and Non-determinism: Exploring power and limitations, conversion between DFA and NFA
Push-Down Automata (PDA): Deterministic vs non-deterministic PDA, equivalence with context-free grammar
Context-Free Grammars (CFGs): Ambiguity, Chomsky normal form, Greibach normal form
Pumping Lemma: Applications to prove non-regular and non-context-free languages
Closure Properties: Regular and context-free languages, advanced examples from real-world applications (e.g., parsing in compilers)
Turing Machines (TM): Variants (multi-tape, non-deterministic, universal TMs), simulation of one machine by another
Translation Between Models: TM vs finite automata vs PDA, equivalence and practical examples (e.g., algorithm simulation)
Model Independence: Robustness of computation models, equivalence of different models of computation
Primitive and Partial Recursive Functions: Definitions, examples, and relation to real-world computations
Encoding Turing Machines: Gödel numbering, self-replication, and encoding schemes
Recursive and Recursively Enumerable Sets: Decidable vs. semi-decidable problems
Rice's Theorem and Recursion Theorem: Proofs, applications, and consequences for software verification
Undecidability and Insolvability: Extensions of the halting problem, practical implications in cryptography and algorithm design
Reducibility: Turing and many-one reducibility, completeness, and real-world undecidable problems
Decidability: Formal language problems, decision problems in software systems (e.g., program termination)
Basic Complexity Classes: Definitions and relationships between P, NP, co-NP, PSPACE, L, NL
Reductions and Completeness: NP-completeness, Cook-Levin theorem, advanced reductions
Hierarchy Theorems: Time and space hierarchy theorems, practical consequences for algorithm efficiency
NP-Completeness and Space-Completeness: Proving NP-hardness, applications in optimization problems
Intractable Problems: Provable limits of computation, approximability, and heuristics for hard problems
P-Completeness and Turing Reductions: Real-world examples (e.g., SAT, 3-SAT), approximation algorithms
Approximation Complexity: Approximation classes (APX), complexity of approximation algorithms
Randomization in Complexity: BPP, RP, and de-randomization techniques in modern computing (e.g., cryptography)

Reference Material:

1. Introduction to Automata Theory, Languages, and Computation" by John E. Hopcroft, Rajeev Motwani, and Jeffrey D. Ullman, Latest Edition.
2. Computability, Complexity, and Languages, by Martin Davis, Ron Sigal, and Elaine J. Weyuker, Latest Edition.
3. Computability and Unsolvability, by Martin Davis, Latest Edition.
4. Introduction to the Theory of Computation, by Michael Sipser, Latest Edition
5. Introduction to Languages and the Theory of Computation, by John C. Martin, Latest Edition
6. Computational Complexity: A Modern Approach, by Sanjeev Arora and Boaz Barak, Latest Edition

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Course Name: Digital Image Processing Using AI
Course Code: CS861
Credit Hours: 3

Course Outline:

Introduction

Fundamentals: Understanding image segmentation and registration, with a focus on medical imaging applications.

Motivation: The significance of accurate segmentation and registration in diagnostics, treatment planning, and image-guided interventions.

Image Segmentation Techniques

Classical Methods: Thresholding, edge detection, and region-based approaches.

ML-Based Approaches: Utilizing Support Vector Machines (SVMs), Random Forests, and K-Nearest Neighbors (KNN) for segmentation tasks.

DL-Based Approaches: Implementing Convolutional Neural Networks (CNNs) and Fully Convolutional Networks (FCNs) for pixel-level classification.

Image Registration Fundamentals

Transformation Models: Exploring rigid, affine, and non-rigid transformations.

Similarity Measures: Techniques such as Mean Squared Error (MSE), Mutual Information, and Cross-Correlation.

Optimization Methods: Strategies including gradient descent and evolutionary algorithms.

Advanced Topics in Registration

Deformable Registration: Addressing challenges in aligning images with complex deformations.

Multimodal Registration: Techniques for registering images from different modalities (e.g., MRI and CT scans).

Real-Time Registration: Approaches to achieve efficient and accurate registration in real-time applications.

Practical Applications

Medical Imaging: Case studies on tumor detection, organ segmentation, and image-guided surgery.

Plant and Fruit Imaging: Applications in disease detection and classification.

Hands-On Projects

Project 1: Implementing a CNN for image segmentation.

Project 2: Developing an algorithm for multimodal image registration.

Project 3: Creating a pipeline for real-time image segmentation and registration.

Reference Materials:

1. *"Deep Learning"* by Ian Goodfellow, Yoshua Bengio, and Aaron Courville: A comprehensive resource on deep learning principles and applications, Latest Edition
2. *"Digital Image Processing"* by Rafael C. Gonzalez and Richard E. Woods: A foundational text covering essential image processing techniques, Latest Edition

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Course Name: MAC Layer Design in Wireless Networks
Course Code: CS862
Credit Hours: 3

Course Outlines:

Introduction to MAC Layer in Wireless Networks; Basic overview, Importance of the MAC layer in wireless networks, Difference between wired and wireless MAC protocols. Challenges in Wireless MAC Design; Hidden and exposed terminal problems, Channel access in shared wireless environments, Wireless spectrum limitations. Classical MAC Protocols; TDMA, FDMA, CDMA, CSMA. Contention-Based MAC Protocols; DCF, EDCA, RTS/CTS, BEB, Adaptive backoff strategies. Reservation-Based MAC Protocols; Scheduling-based MAC protocols, Hybrid MAC protocols. Directional Antenna-Based MAC Protocols; Goals and challenges, Types of directional MAC protocols, Future trends and research directions in directional MAC protocol design. MAC in Multi-Hop and Ad-Hoc Networks; MAC Layer in 4G/5G Networks; LTE MAC protocols, NR MAC, URLLC, mmWave MAC. Energy-Efficient MAC Protocols; Low-duty-cycle MAC protocols, MAC protocols for MANETs. Cognitive MAC Layer Design and Dynamic Spectrum Access; Performance Analysis of MAC Protocols; Throughput, delay and energy consumption analysis, Simulation tools and frameworks. Emerging Trends and Research in MAC Layer Design; MAC protocols for next generation wireless networks, MAC protocols for WBANs, MAC layer Security, Research papers review.

Reference Material:

1. Medium Access Control in Wireless Networks by Yuguang Fang and Yi Pan, Latest Edition
2. Wireless Communications and Networking by William Stallings, Latest Edition
3. Wireless Communications by Andrea Goldsmith, Latest Edition
4. Ad Hoc Wireless Networks: Architectures and Protocols by C. Siva Ram Murthy and B. S. Manoj, Latest Edition

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Course Name: Distributed Agile Software Development
Course Code: CS875
Credit Hours: 3

Course Outline:

Foundations of Agile Software Development, Agile Manifesto, Comparison with traditional software development models, Agile Methodologies and Frameworks (Scrum, Kanban, XP, Lean, etc.), Scaling Agile frameworks: SAFe, LeSS, Nexus, DAD, Advanced backlog management and user story mapping, Techniques for effective Agile estimation and planning, Continuous Integration/Continuous Deployment (CI/CD) in Agile environments, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Acceptance Test-Driven Development (ATDD), Retrospectives, continuous improvement, and innovation in Agile teams, Agile in Distributed and Global Software Development, Managing distributed Agile teams with global tools and technologies, Agile Quality Assurance and DevOps, Agile in AI/ML and Data Science projects, Green and Sustainable Agile practices.

Reference Material:

1. Agile Software Development: The Cooperative Game" by Alistair Cockburn
2. Agile Software Development, Principles, Patterns, and Practices by Robert Martin
3. Agile Software Development with Distributed Teams: Staying Agile in a Global World by Jutta Eckstein
4. "The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations" by Gene Kim, Jez Humble, Patrick Debois, and John Willis

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Course Name: Digital Signal Processing
Course Code: CS780
Credit Hours: 3

Course Outline:

One and N- dimensional signals and systems, Sampling theorem, Discrete-time Fourier transform, discrete Fourier transform, fast Fourier transform, z-transforms: stability and minimum phase signals/systems, Linear filtering of signal: Time domain: Difference equations and convolution, Impulse invariance, bilinear transform, FIR filter design, 2D filter design, Statistical signal processing: Stochastic signals: correlation functions and power density spectra, Optimal filtering: Wiener filters, Adaptive filters: LMS and array processing.

Reference Material:

1. Discrete-Time Signal Processing, Alan V. Oppenheim and Ronald W. Schafer, Prentice-Hall. Latest Edition

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25-Sep-2025
Assistant Registrar
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Course Name: Evolutionary Computing
Course Code: CS781
Credit Hours: 3

Course Outline:

Introduction, Evolutionary Algorithm, Genetic Algorithms, Evolution Strategies, Evolutionary Programming, Genetic Programming, Learning Classifier Systems, Parameter Control in Evolutionary Algorithms, Multi-Modal Problems and Spatial Distribution, Hybridization with other Techniques: Memetic Algorithms, Theory of EAs, Constraint Handling Special Forms of Evolution, Working with Evolutionary Algorithms, Practical implementation of EAs.

Reference Material:

1. Introduction to Evolutionary Computing by Agoston E. Eiben, J.E. Smith. Latest Edition
2. Introduction to Genetic Algorithms by S.N. Sivanandam, S. N. Deepa, Latest Edition
3. Evolutionary Computation by Kenneth A. de De Jong,Kenneth A. De Jong, Latest Edition
4. Genetic Algorithms in Search, Optimization, and Machine Learning David E. Edition Goldberg. Latest
5. Genetic Algorithms + Data Structures = Evolution Programs by ZbigniewMichalewicz, Latest Edition

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25-Sep-2025
Assistant Registrar
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Course Name: Network Security
Course Code: CS751
Credit Hours: 3

Course Outline:

Cryptology and simple cryptosystems; Conventional encryption techniques; Stream and block ciphers; DES; More on Block Ciphers; The Advanced Encryption Standard. Confidentiality & Message authentication: Hash functions; Number theory and algorithm complexity; Public key Encryption. RSA and Discrete Logarithms; Elliptic curves; Digital signatures. Key management schemes; Identification schemes; Dial-up security. E-mail security, PGP, S-MIME; Kerberos and directory authentication. Emerging Internet security standards; SET; SSL and IPsec; VPNs; Firewalls; Viruses; Miscellaneous topics.

Reference Material:

1. W. Stallings, Cryptography and Network Security, Prentice Hall PTR, Upper Saddle River, NJ, Latest Edition
2. Kaufman, R. Perlman, M. Speciner, Network Security: Private Communication in a Public World - Prentice Hall PTR, Upper Saddle River, Latest Edition
3. M. Bishop, Computer Security: Art and Science - Addison-Wesley, Latest Edition
4. Stinson, Cryptography: Theory and Practice, CRC Press, Boca Raton, FL,Latest Edition
5. Richard A. Mollin, An Introduction to Cryptography, Chapman and Hall/CRC, Latest Edition

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25-Sep-2025
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Course Name: Wireless Networks
Course Code: CS752
Credit Hours: 3

Course Outline:

Introduction to Wireless Networks, History, Evolution, and Standards of Wireless Networking, Wireless Links and Network Characteristics, Wireless Network Reference Model, Wireless Networking Issues, Wireless Networking Design and Operation of first, second, and third generation wireless networks, Telecommunication Systems (GSM, GPRS, UMTS), CDMA, IEEE Wireless Data Networks, WLAN: Wi-Fi: 802.11, 802.16, 802.15, Wireless LANs, The 802.11 Architecture, The 802.11 MAC Protocol, Advanced Features in 802.11, Cellular Internet Access, Convergence and interoperability of Wireless System: 4G Wireless Networks, Integration of Ad-Hoc Networks

Reference Material:

1. Wireless and Mobile Networks: Concepts and Protocols, by Sunilkumar S. Manvi , Latest Edition
2. T.S. Rappaport, Wireless Communications: Principles & Practice‖, Latest Edition J. Schiller, Mobile Communications‖, Addison Wesley, Latest Edition
3. Computer Networking: A Top-down Approach Featuring the Internet Book by James F. Kurose, Addison Wesley, Publisher: Addison Wesley Latest Edition

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25-Sep-2025
Assistant Registrar
(Academics)
University of Malakand

Course Name: Wireless Sensor Network Systems
Course Code: CS754
Credit Hours: 3

Course Outline:

Basic Overview of the Sensor Network Systems, Introduction to Wireless Sensor Networks (WSNs), Brief historical survey of WSNs, Examples categories of WSNs applications: Civil and Military domain, Recent Trends and Advancements in WSNs, Sensor-node architecture, Network Architecture & Protocol Stack of WSNs, Unique Characteristics of WSNs, Factors influencing WSN's Design, U, Constrains in WSNs, Network Deployment in WSNs (Structured vs Randomized deployment, Network topology, Mobile deployment), Routing Protocols and Routing Challenges in WSNs, Clustering, Data Aggregation, Fault Management, Scalability and Security in WSNs, Modelling and Simulation Tools for WSNs.

Reference Materials:

1. Akyildiz, I. F. and M. C. Vuran, Wireless Sensor Networks , A John Wiley & Sons Ltd. Latest Edition
2. Sohraby, K., et al.Wireless Sensor Networks: Technology, Protocols, and Applications . Hoboken, New Jersey, John Wiley & Sons, Inc.
3. Karl, H. and A. Willig Protocols and Architectures for Wireless Sensor Networks , John Wiley & Sons, Ltd, West Sussex, England.

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25-Sep-2025
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Course Name: Internet of Things (IoT)
Course Code: CS756
Credit Hours: 3

Course Outlines:

Overview and Basic Concepts of IoT, Architectural Design of IoT, Classification and characteristics of objects, Purpose and applications of IoT, Features of the IoT (Overall aspects, Applications/services aspects, Networking aspects, Link/physical layer aspects, Smart/connected objects aspects, Smart environment aspects), Issues and Challenges in IoT, Identifier for objects and services (devices, tracking, data processing, global connectivity, scalability, autonomics, security, privacy, and web services), Visions of IoT and Goals for new architecture/framework, Cloud Centric IoT concepts, IoT Privacy, Security and Governance, Internet of Things (IoT) Standardization, Status and Requirements, Initiatives and Organizations, Simulation Tools for IoT applications.

Reference Material:

1. Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, by Dr. Ovidiu Vermesan SINTEF, Norway and Dr. Peter Friess EU, Belgium, River Publication Series in Communication by River Publishers, Denmark. Latest Edition
2. Peter Waher, Learning Internet of Things‖, PACKT publishing, BIRMINGHAM – MUMBAI
3. THE INTERNET OF THINGS: AN OVERVIEW Understanding the Issues and Challenges of a More Connected World, By Karen Rose, Scott Eldridge, Lyman Chapin, . The Internet Society. Latest Edition
4. Tutorials, Lectures, and Survey papers

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25-Sep-2025
Assistant Registrar
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Course Name: Distributed Databases
Course Code: CS734
Credit Hours: 3

Course Outline:

Introduction to distributed database systems (DDBMS), architectural models, DDBMS architecture, distributed database design strategies, design issues, fragmentation, allocation, view management, data security, semantic integrity control, distributed query processing problems, query decomposition, localization of distributed data, query optimization, join ordering in fragment queries, distributed query optimization algorithms, transaction processing, concurrency control mechanisms, serializability theory, locked-based and timestamp-based algorithms, optimistic algorithms, deadlock management, reliability concepts and measures, failures in DDBMS, local reliability protocols, distributed reliability protocols, dealing with site failures, Network partitioning, database integration, data processing in multi-databases and inter-operability issues.

Reference Materials:

1. Principles of Distributed Database Systems, M.T. Ozsu, P. Valduriez: Prentice Hall, Latest Edition
2. Transactional Information Systems, G. Weikum and G. Vossen, Morgan Kaufmann, Latest Edition

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